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ENGINEERING SCIENCE

# Topology Optimization of Heat Transfer Applications Operating in the Turbulent Regime

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Thesis submitted for the degree of
Master of Science in Mechanical
Engineering

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# Preface

This thesis marks the culmination of my Master of Science in Mechanical Engineering at KU Leuven. The journey of exploring computational fluid dynamics and mathematical optimization has been both intensely challenging and deeply rewarding. Bridging the gap between theoretical mathematics and practical engineering applications required countless hours of coding, debugging, and analyzing flow fields, but seeing the optimizer successfully carve out organic, aerodynamic fluid channels made the entire process worthwhile.

I would like to express my sincere gratitude to my supervisors, Prof. Niels Horsten and Prof. Maarten Blommaert, for providing me with the opportunity to work on this fascinating topic. Their academic guidance, critical insights, and continuous support were instrumental in shaping the direction and rigor of this research. A very special thanks goes to my assistant-supervisor, Esteban Foronda Obando, for his day-to-day patience, technical expertise, and willingness to help me navigate the complex numerical hurdles within FEniCS.

Furthermore, I would like to thank the assessors, Prof. Elke Deckers and Olivier Renders, for their time and valuable feedback in evaluating this work.

Finally, I want to thank my family and friends for their unwavering encouragement, patience, and perspective throughout the course of my studies.

*Tiebert Lefebure*

# Contents

# Abstract

As modern electronic and automotive systems grow increasingly compact and powerful, efficient thermal management has emerged as a fundamental engineering challenge. While topology optimization has proven to be a highly effective computational design tool for the fluid manifolds that drive these cooling systems, traditional methodologies rely heavily on laminar flow approximations. These laminar assumptions severely limit industrial applicability, as they fail to capture the separation, curvature effects, and turbulence-model dependence inherent to high-speed internal flows.

This thesis develops and implements a density-based topology optimization framework for turbulent internal flows as a baseline for future heat-transfer applications. The computational methodology is built upon the open-source FEniCS platform and utilizes the fictitious domain method with Brinkman momentum penalization to conceptualize solid boundaries within a continuous fluid mesh. To capture the complex physics of high-Reynolds flows, the framework integrates the one-equation Spalart-Allmaras (SA) turbulence model. The optimization process is driven by a frozen-turbulence continuous adjoint, meaning that the eddy-viscosity and wall-distance fields are updated in the forward loop but treated as fixed coefficients during sensitivity assembly.

The assessment strategy is deliberately separated into methodology and results. First, the predictive scope of the SA model for curvature-dominated internal flows is established. The FEniCS-SA implementation is then verified against Ansys Fluent on turbulent channel and U-bend benchmarks. Finally, the pipe-bend and U-bend topology optimization benchmarks from Bayat et al. are evaluated, with the pipe-bend derivative checked through a Dilgen-style finite-difference sensitivity verification before the optimized geometries are exported to Ansys Fluent for sharp-geometry validation with SA and SST $k$-$\omega$ turbulence models. Laminar-versus-turbulent design comparison is retained as supporting context rather than treated as CFD validation in itself.

The resulting thesis is therefore intentionally conservative in its claims. Its central question is not merely whether turbulent topologies look different from laminar ones, but whether SA-driven topology optimization produces low-loss designs whose predicted performance remains credible under independent CFD re-analysis. This establishes a more defensible and transparent baseline for future three-dimensional and conjugate heat-transfer extensions.

# List of Figures and Tables

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# List of Abbreviations and Symbols

## Abbreviations

| | |
|---|---|
| CAD | Computer-Aided Design |
| CEV | Constant Eddy Viscosity |
| CFD | Computational Fluid Dynamics |
| CG1 | Continuous Galerkin (Linear Basis Functions) |
| CHT | Conjugate Heat Transfer |
| DG0 | Discontinuous Galerkin (Piecewise Constant Elements) |
| DNS | Direct Numerical Simulation |
| FEM | Finite Element Method |
| FVM | Finite Volume Method |
| IPCS | Incremental Pressure Correction Scheme |
| MMA | Method of Moving Asymptotes |
| PDE | Partial Differential Equation |
| RAMP | Rational Approximation of Material Properties |
| RANS | Reynolds-Averaged Navier-Stokes |
| SA | Spalart-Allmaras |
| SIMP | Solid Isotropic Material with Penalization |
| SST | Shear Stress Transport |
| TO | Topology Optimization |
| UFL | Unified Form Language |

## Symbols

### Fluid Dynamics & Turbulence

| | |
|---|---|
| $d, y$ | Distance to the nearest wall / Hydraulic diameter [m] |
| $De$ | Dean number [-] |
| $k$ | Turbulent kinetic energy [$m^2/s^2$] |
| $p, \bar{p}, p'$ | Instantaneous, mean, and fluctuating static pressure [Pa] |
| $\Delta P$ | Macroscopic total pressure drop [Pa] |
| $R_c$ | Radius of curvature [m] |
| $Re$ | Bulk Reynolds number [-] |
| $S, \tilde{S}$ | Scalar vorticity magnitude and modified SA vorticity magnitude [1/s] |
| $t$ | Time / Pseudo-time [s] |
| $\mathbf{u}, \bar{u}_i$ | Velocity vector / Mean velocity component [m/s] |
| $u'_i$ | Fluctuating velocity component [m/s] |
| $u_\tau$ | Friction velocity [m/s] |
| $y^+$ | Non-dimensional wall distance [-] |
| $\mu, \mu_t$ | Dynamic and turbulent eddy viscosity [Pa·s] |
| $\nu, \nu_t$ | Kinematic and turbulent eddy kinematic viscosity [$m^2/s$] |
| $\tilde{\nu}$ | Spalart-Allmaras working variable [$m^2/s$] |
| $\rho$ | Fluid density [$kg/m^3$] |
| $\Omega_{ij}$ | Vorticity tensor [1/s] |
| $\kappa$ | von Kármán constant [-] |

### Numerical Implementation (FEniCS)

| | |
|---|---|
| $G, G_0$ | Relaxed Eikonal field variable and boundary value [1/m] |
| $h, h_{avg}$ | Finite element cell diameter and average cell size [m] |
| $\|\| \cdot \|\|_{L_2}$ | $L_2$ (Euclidean) norm [-] |
| $[\![\cdot]\!]$ | Jump operator across an internal element facet [-] |
| $\mathbf{n}$ | Outward-pointing normal vector [-] |
| $v, \xi$ | Test functions for pressure and turbulence equations [-] |
| $\alpha_u, \alpha_{\tilde{\nu}}$ | Relaxation factors for velocity and turbulence [-] |
| $\Delta t$ | Pseudo-time step [s] |
| $\epsilon_{tol}$ | Convergence tolerance [-] |
| $\phi$ | General scalar field variable [-] |
| $\sigma_w$ | Eikonal relaxation constant [-] |

## Topology Optimization

| | |
|---|---|
| $\alpha(\gamma)$ | Continuous Brinkman penalization term / inverse permeability |
| $\alpha_{solid}, \alpha_{fluid}$ | Maximum solid and minimum fluid Brinkman penalties |
| $\alpha_{dg}$ | Artificial diffusion scaling factor for DG0 filter [-] |
| $\beta$ | Heaviside projection steepness parameter [-] |
| $\gamma, \gamma_{raw}$ | Continuous and raw design variable (1 for fluid, 0 for solid) [-] |
| $\tilde{\gamma}, \bar{\gamma}$ | Filtered and Heaviside-projected design variables [-] |
| $\eta$ | Heaviside projection threshold [-] |
| $J$ | Objective function (Power Dissipation) [W] |
| $q$ | RAMP penalization tuning parameter [-] |
| $r$ | Continuous Helmholtz filter radius [m] |
| $\mathbf{s}, \mathbf{s}^*$ | Monolithic forward state and continuous adjoint state vectors |
| $V_f, V_{domain}$ | Physical fluid volume and total design domain volume [$m^3$] |
| $V_{max}$ | Maximum allowable fluid volume fraction constraint [-] |
| $\Omega, \partial\Omega, \Gamma_{int}$ | Computational domain, domain boundary, and internal element facets |

# Chapter 1

# Introduction

## 1.1 Background and Motivation

With the continued miniaturization of components in the electronics and automotive industries, thermal management has become a critical bottleneck in the development of new technologies. While the heat sinks themselves are vital, the efficiency of any cooling system relies heavily on the fluid manifolds that deliver the coolant. If the flow distribution is uneven or if pumping the fluid requires excessive power due to massive pressure drops, the overall efficiency of the system severely degrades. Addressing this fluid delivery problem from an engineering perspective is a challenging task due to the complex aerodynamic behaviors that govern high-speed flow networks.

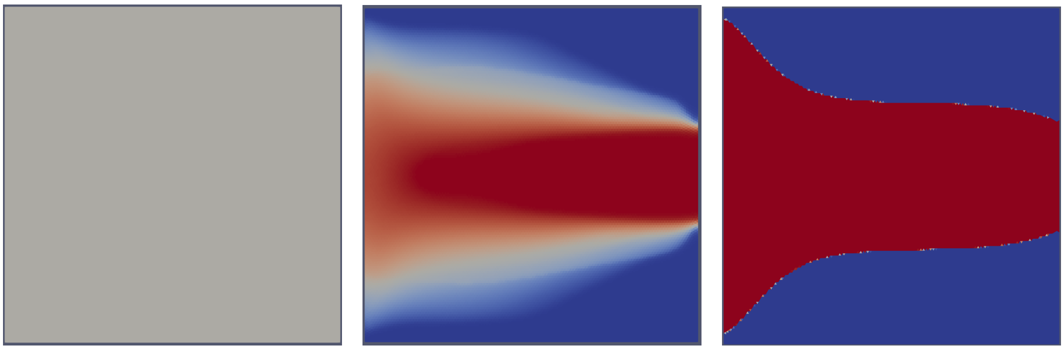
Innovative design strategies based on mathematical optimization have made it possible to improve the performance of these devices while capturing the complexity of the underlying physical phenomena. Among these strategies, topology optimization has emerged as a highly effective computational design tool for advanced thermal management [18]. For instance, the geometry of a complex microfluidic channel network can be automatically optimized to distribute coolant with specified flow rates without relying on human intuition [50]. Furthermore, the ultimate frontier of this technology lies in the design of conjugate and two-fluid heat exchangers, where the algorithm must simultaneously maximize heat transfer while strictly limiting the fluid pumping power [25]. However, before multi-physics heat transfer can be reliably optimized at an industrial scale, the fundamental algorithms governing the turbulent fluid dynamics must first be stabilized and validated.

## 1.2 Conceptual Overview of Density-Based Topology Optimization

To understand the challenges of applying optimization to turbulent fluids, it is first necessary to distinguish topology optimization from traditional design methodologies. Historically, computational design relied on size optimization (altering the thickness of predefined components) or shape optimization (moving the boundary nodes of a predefined geometry). While effective, these methods are inherently limited by their starting topology. A shape optimization algorithm cannot spontaneously create a new hole or split a single fluid channel into two.

Density-based topology optimization, pioneered by Bendsøe and Kikuchi in 1988 for solid mechanics [12], approaches the problem fundamentally differently. Instead of moving boundaries, the design space (the computational domain) is discretized into a fixed grid of finite elements. The algorithm then determines the optimal distribution of material across this grid. In fluid problems, this is achieved by assigning a continuous design variable, $\gamma \in [0, 1]$, to every element in the domain. In the specific convention adopted for this thesis, a value of $\gamma = 1$ represents a pure fluid channel (e.g., the cooling air or water), while $\gamma = 0$ represents a solid, impermeable structure (e.g., a manifold wall).

However, because gradient-based optimizers require continuous and differentiable functions, this discrete binary choice is mathematically relaxed. The design variable is allowed to take intermediate "gray" values between 0 and 1. To give physical meaning to these intermediate states, the fluid equations are modified using a fictitious porous medium approach, where intermediate densities correspond to varying levels of flow permeability. The optimization algorithm iteratively updates the $\gamma$ field (as conceptualized in Figure 1.1) to minimize a specific objective function (e.g., dissipated power) subject to physical constraints (e.g., a maximum allowable fluid volume fraction). The choice of the initial value for $\gamma$ is a critical parameter for solver stability. A uniform initial guess of $\gamma = 0.5$ is standard practice to ensure an unbiased algorithmic search across the design space.



(A) Initial State ($\gamma = 0.5$) (B) Continuous Distribution (C) Discrete Topology

Figure 1.1: Iterative density-based topology optimization process: (a) initialization with a uniform field, (b) intermediate continuous distribution where $\gamma \in [0, 1]$, and (c) final discrete fluid topology where $\gamma \in \{0, 1\}$.

## 1.3 The Evolution of Fluid Topology Optimization

While density-based topology optimization became a standard tool in structural engineering in the late 1990s, its application to fluid dynamics is relatively recent.

The optimization of fluid manifolds remains an active and highly challenging area of research [18]. The genesis of fluid topology optimization is widely attributed to Borrvall and Petersson (2003), who successfully applied the technique to minimize power dissipation in creeping Stokes flow [13]. This was quickly followed by Gersborg-Hansen et al. (2005), who extended the framework to steady, laminar Navier-Stokes flows [20]. Over the past decade, the literature surrounding laminar flow optimization expanded extensively, successfully adapting the methodology to standard incompressible Navier-Stokes equations [15, 3, 50].

However, the assumption of laminar flow significantly simplifies the governing equations and the subsequent derivation of the adjoint sensitivities required for gradient-based optimization. Unfortunately, this assumption prevents the application of these methodologies to most industrial and high-performance applications, where flow velocities are high and turbulence dominates the flow physics. Only recently has the computational mechanics community begun to tackle the severe non-linearities and numerical instabilities associated with optimizing geometries directly in the fully turbulent regime [47, 16, 48].

Extending topology optimization to the turbulent regime is highly non-trivial. It requires coupling optimization algorithms with complex Reynolds-Averaged Navier-Stokes (RANS) turbulence models. These models introduce steep velocity gradients, highly non-linear behaviors, and severe numerical instabilities during the iterative optimization process. To mitigate this, gradient-based turbulent optimizations frequently rely on the "frozen turbulence" assumption, where the turbulent viscosity field is held constant during the sensitivity analysis to avoid differentiating the complex turbulence transport equations. While this assumption provides necessary solver stability and has driven massive success in industrial shape optimization for automotive aerodynamics [30], advancing toward fully coupled turbulent sensitivities in topology optimization remains an ongoing and highly complex mathematical challenge.

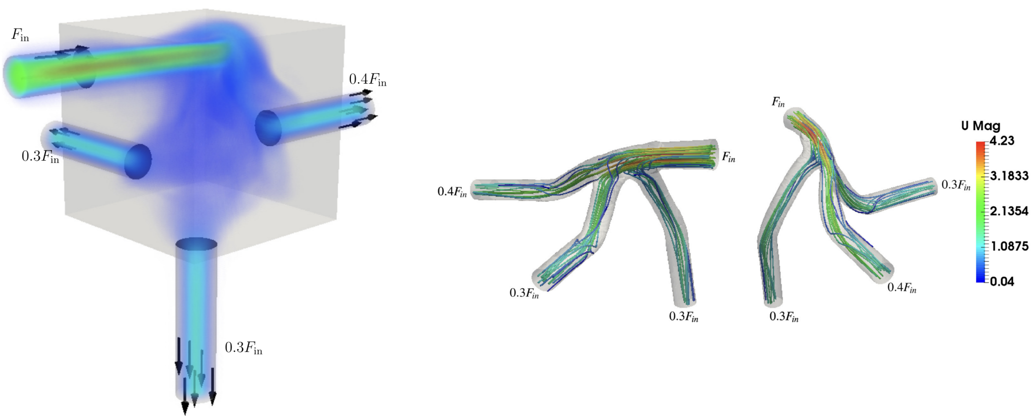



FIGURE 1.2: Topology optimized manifolds with turbulent flow. Adapted from Dilgen et al. [16].

Pioneering works by Yoon (2016) [47] and Dilgen et al. (2018) [16] have pushed the boundaries of these turbulent frameworks. As seen in Figure 1.2, integrating advanced turbulence models allows the optimizer to generate highly complex, organic fluid manifolds specifically tailored to manipulate and guide turbulent flow distributions efficiently.

## 1.4 Objectives and Research Questions

The main objective of this Master's thesis is to extend the application of topology optimization to the design of fluid manifolds operating strictly in the turbulent regime. To achieve this goal, this work investigates the coupling of fluid dynamics and turbulence modeling within an automated mathematical optimization framework. By strictly isolating the fluid mechanics from complex thermodynamic calculations, this thesis bounds its scope to establish a highly stable and mathematically rigorous baseline for pure aerodynamic manifold design.

Rather than relying solely on black-box commercial Computational Fluid Dynamics (CFD) software, the core of this thesis involves implementing the framework using **FEniCS**, an open-source computing platform for solving partial differential equations [4]. FEniCS, combined with high-level discrete adjoint methods (e.g., `dolfin-adjoint` [5]), provides the mathematical flexibility required to implement, verify, and integrate the one-equation Spalart-Allmaras (SA) turbulence model into a fluid-design loop. For full transparency and reproducibility, the complete FEniCS source code developed for this thesis is open-source and publicly available at: `https://github.com/TiebertLefebure/TurbulentTO.git`.

At this stage, it is important to distinguish two different comparisons that appear later in the thesis. A comparison between laminar-optimized and turbulent-optimized layouts is useful as supporting design context, because it shows how inertia and turbulence alter the preferred topology. However, this comparison is not treated as the principal CFD validation step. The decisive assessment instead consists of re-analyzing the final SA-optimized geometries in Ansys Fluent, first with the same SA model and then with alternative RANS closures.

Based on this methodology, this thesis seeks to answer the following specific research questions:

1. **How accurately does the Spalart-Allmaras turbulence model predict curvature-dominated internal turbulent flows, and where do its physical limitations appear?**

2. **Can a custom standard-Galerkin FEniCS implementation of the Spalart-Allmaras turbulence model converge robustly and achieve quantitative agreement with established commercial CFD solvers?**

3. **To what extent do SA-driven topology optimizations produce final designs whose predicted performance remains consistent under independent Ansys re-analysis, both with the same SA model and with alternative RANS closures?**

## 1.5 Outline of the Thesis

The remainder of this thesis is structured into the following six chapters:

- **Chapter 2 (Turbulence Modeling):** Provides the theoretical background on fluid dynamics and turbulence. It discusses the Navier-Stokes equations, the principles of RANS modeling, and the physical characteristics of the Spalart-Allmaras model.
- **Chapter 3 (Implementation of the Spalart-Allmaras Model in FEniCS):** Details the numerical implementation of the chosen turbulence model within the FEniCS framework, including fractional-step solvers, standard Galerkin discretization, Picard coupling, and wall-distance computation.
- **Chapter 4 (Turbulent Topology Optimization Methodology):** Introduces the density-based optimization framework. It details the Brinkman momentum penalization, the power-law SA and wall-distance penalties, the reciprocal penalized wall-distance equation, the pressure objective, and the rationale for adopting the frozen turbulence assumption as the baseline design engine.
- **Chapter 5 (Verification of the Spalart-Allmaras Solver for Turbulent Internal Flows):** Verifies the standalone FEniCS-SA forward solver against Ansys Fluent on turbulent channel-flow and U-bend benchmarks.
- **Chapter 6 (Turbulent Topology Optimization Results):** Presents the turbulent topology optimization results for the pipe-bend and U-bend benchmarks from Bayat et al., documents the pipe-bend sensitivity verification, exports the final designs to Ansys Fluent, and evaluates the designs with both SA and SST $k$-$\omega$ turbulence models.
- **Chapter 7 (Conclusion and Future Work):** Summarizes the main findings, answers the research questions conservatively, and outlines the remaining limitations, with particular emphasis on turbulence-model dependence, geometry extraction, and future multi-physics heat-transfer applications.

# Chapter 2

# Turbulence Modeling

## 2.1 Introduction to Fluid Dynamics and Turbulence

Fluid dynamics is governed by the Navier-Stokes equations, which describe the conservation of mass and momentum for a viscous continuum fluid [45]. In this introductory text, these equations are presented using index notation (Einstein summation convention) to clearly illustrate the individual tensor components. In this convention, the indices $i$ and $j$ represent the three spatial dimensions (1, 2, and 3, corresponding to the $x$, $y$, and $z$ axes). Consequently, $x_i$ denotes the Cartesian coordinates, and $u_i$ represents the corresponding components of the velocity vector. A repeated index within a single term implies a mathematical summation over all three spatial directions. (Note: In subsequent chapters dealing with finite element implementation, this notation will shift to standard vector calculus to align with FEniCS unified form language syntax). The fundamental principle of mass conservation is expressed by the continuity equation. For a general, compressible fluid, this is defined as:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_i)}{\partial x_i} = 0 \tag{2.1}$$

where $\rho$ is the fluid density. However, for most liquid cooling applications and low-speed fluid manifolds, the fluid can be safely assumed to be incompressible. Under the incompressible flow assumption, the density $\rho$ is constant in both space and time. Therefore, the time derivative of density becomes zero, and the constant density can be factored out of the spatial derivative. Dividing the equation by $\rho$ yields the simplified, incompressible continuity equation:

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{2.2}$$

Similarly, for an incompressible, Newtonian fluid with a constant dynamic viscosity $\mu$, the conservation of momentum is expressed as:

$$\rho\left(\frac{\partial u_i}{\partial t} + u_j\frac{\partial u_i}{\partial x_j}\right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\mu\frac{\partial u_i}{\partial x_j}\right) \tag{2.3}$$

where $p$ represents the static pressure of the fluid. In principle, solving Equation 2.3 directly yields the exact flow field. This approach is known as Direct Numerical Simulation (DNS). However, as the Reynolds number ($Re$) increases, the flow transitions into a chaotic turbulent state characterized by a broad spectrum of spatial and temporal scales [32]. Resolving the smallest turbulent eddies requires extremely fine spatial grids. For industrial applications, including the design of electronics cooling manifolds, DNS is computationally prohibitive. Therefore, engineers rely on turbulence modeling to predict the averaged effects of these fluctuations without resolving the complete turbulent spectrum [43, 6].

## 2.2 Reynolds-Averaged Navier-Stokes (RANS) Equations

The most common approach to turbulence modeling in engineering is the Reynolds-Averaged Navier-Stokes (RANS) framework, based on the theories originally introduced by Osborne Reynolds [32]. The core principle of RANS is the Reynolds decomposition. This mathematical operation separates the instantaneous flow variables into a time-averaged mean component ($\bar{u}_i$, $\bar{p}$) and a fluctuating component ($u'_i$, $p'$):

$$u_i = \bar{u}_i + u'_i \quad \text{and} \quad p = \bar{p} + p' \tag{2.4}$$

Substituting these decomposed variables into the standard Navier-Stokes equations and taking the time average yields the RANS momentum equations:

$$\rho \left( \frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} \right) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left( \mu \frac{\partial \bar{u}_i}{\partial x_j} - \rho \overline{u'_i u'_j} \right) \tag{2.5}$$

The averaging process introduces a new term, $-\rho \overline{u'_i u'_j}$, known as the Reynolds stress tensor. This tensor represents the momentum transfer caused by turbulent fluctuations. The introduction of this term creates a fundamental mathematical closure problem because the RANS equations now contain more unknowns than available physical equations [43].

### 2.2.1 The Boussinesq Hypothesis and the Isotropic Assumption

To close the system and solve for the mean flow, most RANS models rely on the Boussinesq eddy viscosity hypothesis [32]. This hypothesis assumes that the turbulent momentum transfer is analogous to molecular momentum transfer. It relates the Reynolds stresses linearly to the mean velocity gradients using a proportionality constant called the turbulent eddy viscosity ($\mu_t$):

$$-\rho \overline{u'_i u'_j} = \mu_t \left( \frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \frac{2}{3} \rho k \delta_{ij} \tag{2.6}$$

where $k$ is the turbulent kinetic energy and $\delta_{ij}$ is the Kronecker delta. The primary task of any Boussinesq-based turbulence model is to calculate this local eddy viscosity. It is crucial to highlight a fundamental physical limitation of Equation 2.6: the Boussinesq hypothesis inherently assumes that $\mu_t$ is an *isotropic* scalar quantity. It mathematically enforces that turbulence behaves identically in all spatial directions and aligns perfectly with the mean strain rate. While this assumption holds for simple shear flows, it systematically fails in predicting anisotropic phenomena, such as the strong secondary flows induced by streamline curvature.

## 2.3 Boundary Layer Theory and Near-Wall Treatment

Before selecting a specific turbulence equation to calculate $\mu_t$, the physical structure of a turbulent boundary layer must be considered, as the model must accurately resolve these near-wall regions. The boundary layer is traditionally divided into three distinct regions based on the non-dimensional wall distance ($y^+$) and non-dimensional velocity ($u^+$):

$$y^+ = \frac{y u_\tau}{\nu}, \quad u^+ = \frac{\bar{u}}{u_\tau} \tag{2.7}$$

where $y$ is the absolute distance from the wall, $\nu$ is the kinematic viscosity ($\mu/\rho$), and $u_\tau = \sqrt{\tau_w/\rho}$ is the friction velocity derived from the wall shear stress ($\tau_w$) [32].

1. **The viscous sublayer ($y^+ < 5$):** Immediately adjacent to the wall, turbulent fluctuations are damped out by kinematic constraints. Viscous shear dominates, and the velocity profile is strictly linear ($u^+ = y^+$).

2. **The buffer layer ($5 < y^+ < 30$):** A transition region where both viscous and turbulent shear stresses are of equal magnitude.

3. **The log-law region ($y^+ > 30$):** Farther from the wall, turbulence dominates, and the velocity follows a logarithmic distribution predicted by the von Kármán constant [43].

High-Reynolds models (like the standard $k-\epsilon$ model) use empirical "wall functions" to bridge the gap between the wall and the log-law region, requiring the first computational mesh element to be placed at $y^+ > 30$. However, for topology optimization, where solid-fluid boundaries continuously evolve during the optimization loop, empirical wall functions introduce severe mathematical discontinuities and numerical instabilities. Therefore, a "wall-resolved" model that integrates the governing equations all the way through the viscous sublayer ($y^+ \approx 1$) is strictly required for robust gradient-based optimization.

## 2.4 The Spalart-Allmaras One-Equation Model

While two-equation models (such as $k-\epsilon$ or $k-\omega$) solve separate transport equations for the turbulent kinetic energy and a turbulent length scale, they introduce highly non-linear source terms. These terms frequently cause numerical instabilities during the adjoint sensitivity analysis required for topology optimization. To achieve a trade-off between physical accuracy and computational efficiency, this thesis utilizes the Spalart-Allmaras (SA) model, originally developed in 1992 for aerodynamic flows with adverse pressure gradients [35]. The SA model is a one-equation model that solves a single transport equation directly for a working variable, $\tilde{\nu}$, which is subsequently linked to the true kinematic eddy viscosity ($\nu_t = \mu_t/\rho$) [36]. The transport equation for the standard SA model is given by:

$$\frac{\partial \tilde{\nu}}{\partial t} + \bar{u}_j \frac{\partial \tilde{\nu}}{\partial x_j} = c_{b1} \tilde{S} \tilde{\nu} - c_{w1} f_w \left( \frac{\tilde{\nu}}{d} \right)^2 + \frac{1}{\sigma} \left[ \frac{\partial}{\partial x_j} \left( (\nu + \tilde{\nu}) \frac{\partial \tilde{\nu}}{\partial x_j} \right) + c_{b2} \left( \frac{\partial \tilde{\nu}}{\partial x_j} \right)^2 \right] \tag{2.8}$$

The terms on the right-hand side represent the production, destruction, and diffusion of the turbulent working variable, respectively. Crucially, the production term relies on a modified vorticity scalar, $\tilde{S}$. This indicates that turbulence generation in the SA model is fundamentally driven by fluid shear, a physical characteristic that introduces unique boundary-condition sensitivities during advanced adjoint calculations (discussed further in Chapter 4). Furthermore, the destruction term is explicitly a function of $d$, the absolute distance to the nearest wall. This makes the SA model a highly effective "wall-resolved" low-Reynolds model. It is inherently sensitive to near-wall damping effects and integrates smoothly through the viscous sublayer without requiring complex empirical wall functions or a second length-scale equation [33]. Detailed derivations of the model constants and wall damping functions are deferred to Chapter 3, where the numerical implementation in FEniCS is discussed.

## 2.5 Evaluation of Curvature Effects and the Dean Number

Despite its computational advantages for topology optimization, the standard SA model relies on a single length scale tied to the nearest wall distance. As it lacks an explicit rotation/curvature correction and remains constrained by the Boussinesq eddy-viscosity assumption, it struggles to capture the anisotropic physics of streamline curvature. When fluid passes through a bend, centrifugal forces induce secondary flows perpendicular to the main flow direction. To quantify the strength of these secondary flows, the dimensionless Dean number ($De$) is utilized:

$$De = Re_d \sqrt{\frac{d}{2R_c}} \tag{2.9}$$

where $d$ is the characteristic length of the cross-section (e.g., the pipe diameter $D$ or hydraulic diameter $D_h$), $R_c$ is the radius of curvature, and $Re_d$ is the bulk Reynolds number based strictly on that exact same characteristic length. A higher Dean number indicates a stronger cross-sectional influence of centrifugal forces.

A review of the literature demonstrates that the predictive capability of the SA model is heavily dependent on this Dean number. For mildly curved flows, the SA model performs relatively well. Apalowo [8] evaluated a 90° circular pipe bend with a Dean number of $De \approx 9,000$ (operating at $Re_D \approx 34,000$, based on the pipe diameter). Under these moderate conditions, where the bend radius was seven times larger than the pipe diameter ($R_c = 7D$), the SA model provided a highly acceptable approximation of the streamwise velocity profiles compared to higher-order models, successfully capturing the primary flow physics.

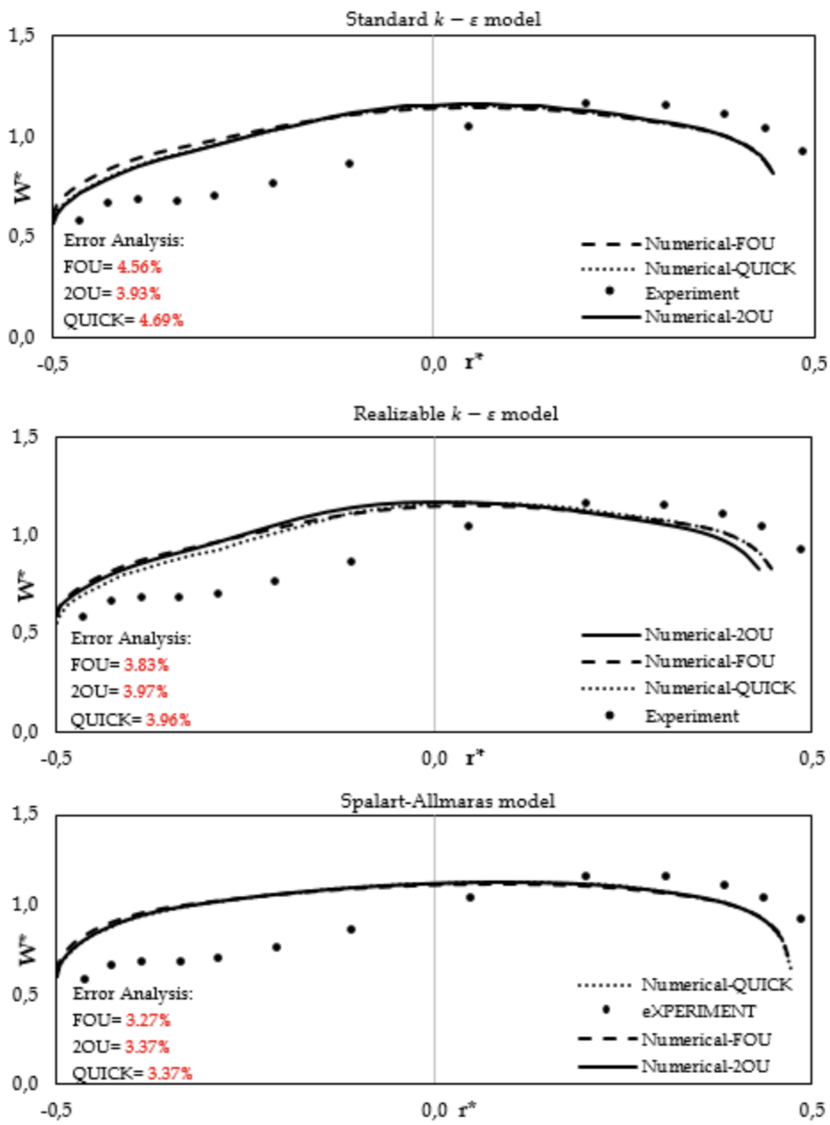



Figure 2.1: Streamwise velocity profiles in a 90° pipe bend ($R_c = 7D$, $Re_D \approx 34,000$, $De \approx 9,000$). The SA model performs relatively well, closely matching the experimental data and two-equation models for mild curvature. Reprinted from Apalowo [8].

However, as the curvature tightens and the Dean number increases, the limitations of the isotropic Boussinesq assumption become severe. Abuhatira et al. [1] evaluated the internal turbulent flow through a 90° pipe bend characterized by a tighter curvature ratio of $R_c = 2.0D$. Operating at a bulk Reynolds number of $Re_D = 60,000$, this geometry generated strong centrifugal forces and a corresponding Dean number of $De \approx 30,000$. Unlike extreme benchmark cases where all isotropic models fail uniformly, this configuration distinctly isolates the shortcomings of the Spalart-Allmaras formulation. At the critical bend exit ($X/D = 0$), the SA model drastically mischaracterized the near-wall velocity profiles, completely missing the flow physics close to the inner wall that higher-order two-equation models (such as $k-\epsilon$ and $k-\omega$) resolved with much greater accuracy. This confirms that the failure is not attributed to inertial scaling, but is strictly driven by the model's inability to compensate for strong anisotropic centrifugal effects.

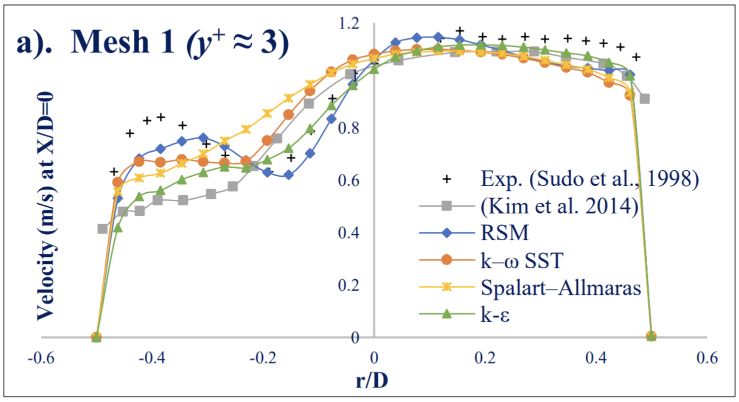



FIGURE 2.2: Turbulence model predictions at the 90° bend exit ($X/D = 0$, $R_c = 2.0D$, $Re_D = 60,000$, $De \approx 30,000$). Due to the strong curvature, the standard Spalart-Allmaras model fails to accurately capture the near-wall velocity profile compared to two-equation closures. Adapted from Abuhatira et al. [1].

While the literature clearly establishes that the SA model is viable at $De \approx 9,000$ and fails to capture inner-wall physics at $De \approx 30,000$, a bracketed or case-specific threshold between these states must be estimated to justify its use in topology optimization. Therefore, to better evaluate where the model begins to break down, rigorous benchmark studies were subsequently conducted within Ansys Fluent.

## 2.6 Benchmark Studies in Ansys Fluent

To investigate where the SA model's accuracy deteriorates, benchmark simulations were performed using **Ansys Fluent**. The study focused on two distinct, fully three-dimensional (3D) internal flow geometries: a 180° U-bend and a tighter 90° pipe bend. To ensure a standardized and rigorously tested numerical baseline, the computational meshes, fluid properties, and boundary conditions for these simulations were adopted directly from the official Ansys Fluid Dynamics Verification

Manual [7]. Specifically, the 90° bend corresponds to verification case VMFL030, while the 3D U-bend corresponds to case VMFL048. These cases evaluate industrial flow regimes ($Re_D > 40,000$) to ensure the turbulence model is rigorously verified against dominant inertial physics before being implemented into the custom FEniCS solver. To evaluate the accuracy of the one-equation model, the results were compared directly against the $k-\omega$ SST model, an industry standard for accurately predicting adverse pressure gradients and flow separation.

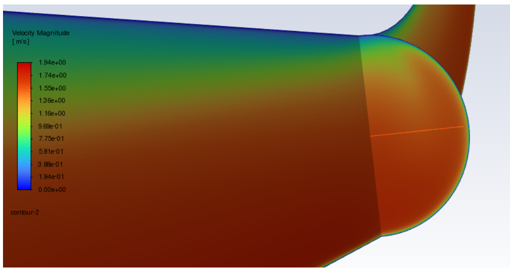
### 2.6.1 Mild Curvature: The U-Bend Case

The first benchmark evaluated a 3D U-bend geometry characterized by a moderate curvature ratio of $R_c \approx 4.46D$. Operating at a bulk Reynolds number of $Re_D \approx 44,700$, this geometry resulted in a Dean number of $De \approx 15,000$. Consistent with the findings of Apalowo for moderate bends, the SA model performed well under these conditions. As shown in Figure 2.3, the qualitative velocity contours of the SA model accurately replicate the bulk flow distribution of the $k-\omega$ SST model. Furthermore, the quantitative velocity profiles agree well with the experimental data from Takamasa and Kondo [40].

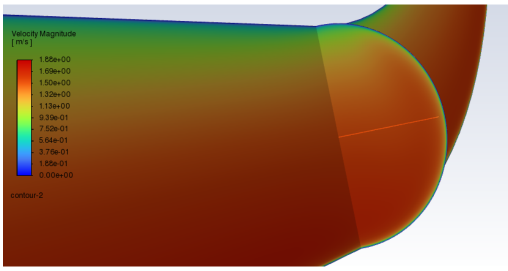
### 2.6.2 Strong Curvature: The 90° Bend Case

The second benchmark evaluated a tighter 90° pipe bend characterized by a severe curvature ratio of $R_c = 2.8D$. Despite operating at a very similar bulk Reynolds number of $Re_D \approx 43,000$ compared to the U-bend, this significantly tighter bend radius pushed the Dean number up to $De \approx 18,000$. Because the bulk flow speed and Reynolds number remain effectively constant between these two benchmark cases, this comparison successfully isolates the geometric effect of streamline curvature.

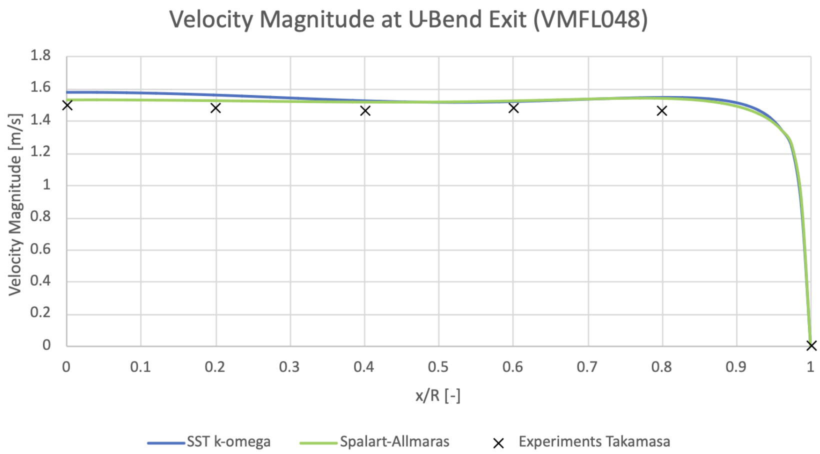
As shown in Figure 2.4, examining the quantitative velocity profile at the bend exit ($\theta = 90°$) clearly demonstrates the breakdown of the SA model. Driven primarily by the tighter bend radius and amplified anisotropic centrifugal effects, the SA model over-predicts the turbulent viscosity. This causes it to over-smooth the velocity profile, failing to capture the sharp velocity gradients near the inner wall ($y/D = 0$) when compared against both the $k-\omega$ SST prediction and the experimental data from Enayet et al. [17]. This failure is also visually apparent when comparing the side-by-side velocity contours.



(A) $k-\omega$ SST Contour

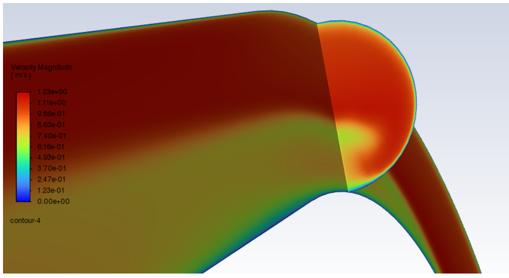


(B) Spalart-Allmaras Contour

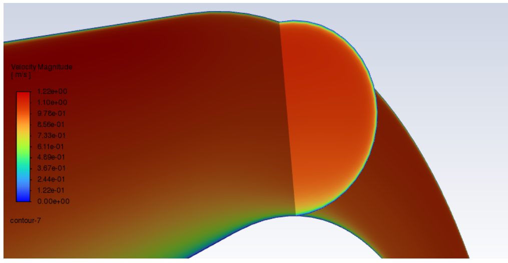



(C) Velocity Profiles

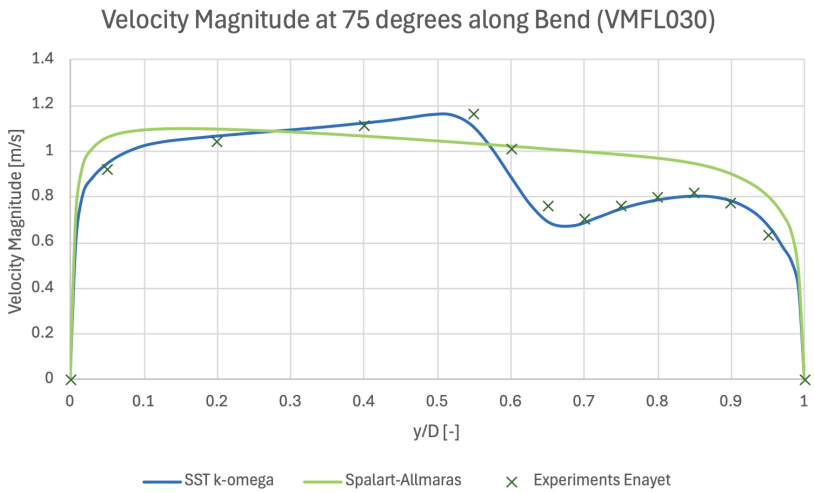
FIGURE 2.3: Flow predictions for a 180° U-bend ($R_c \approx 4.46D$, $Re_D \approx 44,700$, $De \approx 15,000$). The SA model accurately matches the higher-fidelity $k-\omega$ SST contours and directly aligns with the experimental data from Takamasa and Kondo [40].



(A) $k - \omega$ SST Contour



(B) Spalart-Allmaras Contour




(C) Velocity Profiles

Figure 2.4: Flow predictions at the 90° pipe bend exit ($R_c = 2.8D$, $Re_D \approx 43,000$, $De \approx 18,000$). The SA model over-smooths the velocity gradients near the inner wall, deviating significantly from both the $k - \omega$ SST model and the experimental data from Enayet et al. [17].

## 2.7 Conclusion

This chapter directly addresses the first research question regarding the predictive capabilities and physical limitations of the Spalart-Allmaras model. The theoretical breakdown and Ansys benchmark studies confirm that for internal geometries with moderate curvature (up to $De \approx 15,000$), the SA model performs accurately and successfully captures the primary flow physics. Furthermore, because it is a wall-resolved model, it avoids the numerical pitfalls of empirical wall functions, making it exceptionally well-suited for boundary-evolving topology optimization. However, its physical limitation lies in the isotropic Boussinesq assumption. When streamline curvature induces strong anisotropic secondary flows (heuristically observed to cause model breakdown in this case beyond a threshold of $De \approx 18,000$), the uncompensated SA model systematically overpredicts eddy viscosity and fails to capture flow separation. Consequently, while the SA model provides the necessary numerical stability to drive the optimization algorithms in FEniCS, the final topological designs must be critically verified. Any complex manifold generated by the SA optimizer must be subsequently evaluated in commercial software using a higher-fidelity two-equation model to confirm its true aerodynamic performance.

# Chapter 3

# Implementation of the Spalart-Allmaras Model in FEniCS

## 3.1 Introduction

With the theoretical foundation of the Spalart-Allmaras (SA) model established in Chapter 2, this chapter details its numerical translation into a working FEniCS solver. The focus is methodological: the finite-element weak forms, Picard coupling strategy, pressure-correction flow solver, SA transport equation, and relaxed Eikonal wall-distance calculation are described before the solver is verified separately in Chapter 5.

The numerical simulations and eventual topology optimization in this thesis are built upon FEniCS, an open-source computing platform for solving partial differential equations (PDEs). Unlike commercial computational fluid dynamics (CFD) software such as Ansys, which typically relies on the Finite Volume Method (FVM), FEniCS uses the Finite Element Method (FEM) and requires the user to explicitly define the mathematical formulation of the problem [28].

To solve a PDE in FEniCS, the governing equations (the "strong form") must be mathematically transformed into a "weak formulation" (or variational form). FEniCS utilizes the Unified Form Language (UFL) to express these weak forms in Python code that closely mirrors the mathematical notation [26]. This high-level mathematical scripting is particularly advantageous for topology optimization, as it allows for the automated derivation of the adjoint equations required for gradient-based optimization algorithms.

To ensure complete transparency and reproducibility, the entire custom codebase developed for this thesis is available in the public GitHub repository `TurbulentTO`. To maintain modularity, the repository is divided into two primary directories. The `TurbulenceModels` directory contains the standalone Spalart-Allmaras fluid dynamics engine and the verification benchmarks reported in Chapter 5. The `FluidTO` directory houses the optimization algorithms, penalty formulations, and objective functions detailed in Chapter 4.

### 3.1.1 Derivation of a Weak Formulation

Before diving into the complex turbulence equations, it is instructive to demonstrate how a continuous PDE is translated into a discrete FEniCS formulation. To illustrate this methodology, consider the Poisson equation for pressure, which is a fundamental component of incompressible fluid solvers. The strong form of the Poisson equation is given by:

$$-\nabla^2 p = f \quad \text{in}\,\Omega \tag{3.1}$$

where $p$ is the pressure field, $f$ is a source term (derived from the velocity divergence), and $\Omega$ is the computational domain. To obtain the weak form, the equation is multiplied by a sufficiently smooth test function $v$ and integrated over the entire domain $\Omega$, following standard Galerkin finite element procedures [28, 26]:

$$-\int_\Omega (\nabla^2 p) v \, dx = \int_\Omega f v \, dx \tag{3.2}$$

Applying integration by parts (Green's first identity) to the left-hand side yields:

$$\int_\Omega \nabla p \cdot \nabla v \, dx - \int_{\partial\Omega} (\nabla p \cdot \mathbf{n}) v \, ds = \int_\Omega f v \, dx \tag{3.3}$$

where $\partial\Omega$ is the boundary of the domain and $\mathbf{n}$ is the outward-pointing normal vector. This integral formulation fundamentally lowers the continuity requirements of the solution, requiring only first-order derivatives to be square-integrable. Furthermore, Neumann boundary conditions (e.g., a prescribed pressure gradient at an outlet) can be naturally substituted directly into the boundary integral $\int_{\partial\Omega} (\nabla p \cdot \mathbf{n}) v \, ds$. This variational approach forms the foundation of all fluid and turbulence equations implemented in this thesis.

## 3.2 Numerical Solution Strategy for Fluid Dynamics

Solving the full Reynolds-Averaged Navier-Stokes (RANS) equations is numerically challenging due to the non-linear convective terms, the saddle-point nature of the coupled velocity-pressure system, and the intense cross-coupling with the turbulence model. To achieve a stable and efficient steady-state solution in FEniCS that seamlessly translates into the topology optimization framework, the governing equations are rigorously decoupled.

### 3.2.1 Decoupling the Physics: The Picard Iteration Scheme

Because the momentum equation relies on the turbulent eddy viscosity ($\nu_t$), and the turbulence transport equation relies on the convective velocity ($\mathbf{u}$), solving the entire fluid-turbulence system simultaneously in a massive monolithic matrix frequently leads to ill-conditioning and solver divergence.

To resolve this, the forward physics are decoupled using a **Picard iteration scheme** (successive substitution), a robust approach for segregated RANS solvers [19, 43]. During each Picard step, the flow solver advances using the eddy viscosity from the previous iteration. The newly updated velocity field is then passed to the Spalart-Allmaras solver to calculate a new eddy viscosity field. This alternating sequence forms the outer loop of the solver, iteratively updating the decoupled equations until the fully coupled physical system reaches a steady-state equilibrium.

### 3.2.2 Incremental Pressure Correction Scheme (IPCS)

Within the Picard loop, the Navier-Stokes equations themselves must be solved. Rather than solving for velocity and pressure simultaneously, this implementation utilizes an Incremental Pressure Correction Scheme (IPCS). This algorithmic splitting is fundamentally based on the fractional-step projection methods originally pioneered by Chorin [14] and Temam [41]. To nurse the non-linear convective momentum terms to a stable steady state, a pseudo-time stepping approach ($\Delta t$) is employed strictly for the flow variables. The IPCS algorithm splits the fluid equations into three sequential steps per pseudo-time step:

1. **Tentative velocity step:** The momentum equation is solved using the pressure field from the previous iteration ($p^n$) to compute an intermediate, non-divergence-free velocity field ($\mathbf{u}^*$). Because this is a RANS framework, the diffusion term utilizes the effective kinematic viscosity, ($\nu + \nu_t$). Treating the convective term explicitly and the diffusive term implicitly, the semi-discrete equation is:

$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} + (\mathbf{u}^n \cdot \nabla)\mathbf{u}^* - \nabla \cdot ((\nu + \nu_t^n)\nabla \mathbf{u}^*) = -\nabla p^n \tag{3.4}$$

2. **Pressure correction step:** A Poisson equation is solved to find the pressure variation that enforces the continuity constraint ($\nabla \cdot \mathbf{u}^{n+1} = 0$). The equation for the new pressure field ($p^{n+1}$) is derived as:

$$\nabla^2 p^{n+1} = \nabla^2 p^n + \frac{1}{\Delta t}\nabla \cdot \mathbf{u}^* \tag{3.5}$$

3. **Velocity update step:** The tentative velocity is strictly corrected using the gradient of the pressure difference to yield a mass-conserving velocity field ($\mathbf{u}^{n+1}$) for the current iteration:

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla (p^{n+1} - p^n) \tag{3.6}$$

The simulation is deemed fully converged to a steady state when the $L_2$ norm of the residual difference between successive Picard iterations falls below a defined tolerance limit ($\epsilon_{tol}$) for both the fluid and turbulence fields.

## 3.3 Detailed Formulation of the Spalart-Allmaras Model

As concluded in Chapter 2, the Spalart-Allmaras (SA) one-equation model was selected for its balance of physical accuracy and numerical stability. Unlike the transient flow equations, the SA transport equation is solved in its purely steady-state form within the Picard loop. The steady transport equation for the continuous nonlinear working variable $\tilde{\nu}$ is given by:

$$\mathbf{u} \cdot \nabla\tilde{\nu} = c_{b1}\tilde{S}\tilde{\nu} - c_{w1}f_w\left(\frac{\tilde{\nu}}{y}\right)^2 + \frac{1}{\sigma}\left[\nabla \cdot ((\nu + \tilde{\nu})\nabla\tilde{\nu}) + c_{b2}(\nabla\tilde{\nu})^2\right] \tag{3.7}$$

where $\nu$ is the molecular kinematic viscosity, $\sigma = 2/3$ is the turbulent Prandtl number, $c_{b1}$ and $c_{w1}$ are calibrated closure constants, $\tilde{S}$ is a modified vorticity magnitude, $f_w$ is the wall-damping function, and $y$ is the physical distance to the nearest solid boundary [36].

### 3.3.1 Closure Terms and Calibration Constants

The Spalart-Allmaras model relies on several interconnected closure functions to accurately capture boundary layer physics. The production term relies on a modified vorticity magnitude, $\tilde{S}$, which prevents the destruction of turbulence in regions where the vorticity goes to zero, while maintaining correct scaling near the wall [36]. It is defined as:

$$\tilde{S} = S + \frac{\tilde{\nu}}{\kappa^2 y^2} f_{v2} \tag{3.8}$$

where $S = \sqrt{2\Omega_{ij}\Omega_{ij}}$ is the scalar magnitude of the vorticity tensor, $\kappa$ is the von Kármán constant, and $y$ is the distance to the nearest wall. The function $f_{v2}$ is a near-wall modifier:

$$f_{v2} = 1 - \frac{\chi}{1 + \chi f_{v1}} \quad \text{where} \quad \chi = \frac{\tilde{\nu}}{\nu} \quad \text{and} \quad f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3} \tag{3.9}$$

Similarly, the wall destruction term is modulated by the non-dimensional function $f_w$, which accelerates the decay of turbulent viscosity inside the viscous sublayer:

$$f_w = g\left(\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6}\right)^{1/6} \tag{3.10}$$

where the intermediate variables $g$ and $r$ are evaluated as:

$$g = r + c_{w2}(r^6 - r) \quad \text{and} \quad r = \min\left(\frac{\tilde{\nu}}{\tilde{S}\kappa^2 y^2}, 10\right) \tag{3.11}$$

The upper bound of 10 on $r$ ensures numerical stability during solver initialization when $\tilde{S}$ may momentarily approach zero.

It should be explicitly noted that the original formulation by Spalart and Allmaras [36] includes additional geometric trip functions ($f_{t1}$ and $f_{t2}$) designed to manually trigger the transition from laminar to turbulent flow at specific user-defined coordinates. However, following the topology optimization methodology established by Yoon [47], these transition functions are deliberately omitted from this thesis. Because the topological boundaries are unknown *a priori* and evolve during optimization, tracking discrete trip points is mathematically unfeasible; therefore, the model assumes a fully turbulent flow regime.

To close the system, the standard empirical calibration constants formulated by Spalart and Allmaras are utilized throughout the FEniCS implementation [33]. Notably, the von Kármán constant is set to 0.41 to perfectly align with the standard OpenFOAM and generalized industrial implementations[cite: 3]:

$$\begin{aligned} c_{b1} &= 0.1355 & \sigma &= 2/3 & c_{b2} &= 0.622 \\ \kappa &= 0.41 & c_{w2} &= 0.3 & c_{w3} &= 2.0 \\ c_{v1} &= 7.1 & c_{w1} &= \frac{c_{b1}}{\kappa^2} + \frac{1+c_{b2}}{\sigma} \end{aligned}$$

### 3.3.2 Wall-Distance Calculation via the Relaxed Eikonal Equation

A major computational challenge in the SA model is the calculation of the wall distance $y$. In a standard static CFD simulation, $y$ is purely geometric and can be pre-computed. However, to ensure this verification solver translates seamlessly into the topology optimization code (where solid boundaries evolve continuously), the implementation abandons discrete geometric searches in favor of a PDE-based approach.

The true geometric wall distance satisfies the standard Eikonal equation, $|\nabla y| = 1$. However, this formulation contains singularities at the medial axes of the domain. To guarantee continuous differentiability, a requirement for later adjoint derivations, a relaxed Eikonal equation proposed by Yoon [47] is solved:

$$\nabla G \cdot \nabla G + \sigma_w G(\nabla \cdot \nabla G) = (1 + 2\sigma_w)G^4 \tag{3.12}$$

Here, $G$ is an intermediate, non-dimensional field variable. The parameter $\sigma_w$ is a relaxation constant (typically set to 0.1) that controls the smoothness of the field. The PDE is subjected to a Dirichlet boundary condition at the physical walls, $G = G_0$, where $G_0$ is an arbitrarily large constant (e.g., $G_0 = 20.0$). Once the field $G$ is solved over the domain, the physical wall distance $y$ is recovered algebraically at every node using the inverse relationship:

$$y = \frac{1}{G} - \frac{1}{G_0} \tag{3.13}$$

This system is implemented as a nonlinear weak residual for $G$. After multiplying by a scalar test function and integrating the $\sigma_w G \nabla^2 G$ term by parts, the residual contains only first derivatives of $G$. Listing 3.1 shows the weak-form structure used by the solver: the first two terms represent the relaxed Eikonal operator, the third term is the nonlinear source, and the Dirichlet condition fixes $G$ on physical walls before recovering $y$ algebraically.

```
# Unknown reciprocal wall-distance field G, test function z.
F_G = (
    (1.0 - sigma_w) * inner(grad(G), grad(G)) * z * dx
    - sigma_w * G * inner(grad(G), grad(z)) * dx
    - (1.0 + 2.0 * sigma_w) * G**4 * z * dx
)

wall_bc = DirichletBC(V_G, G0, physical_walls)
solve(F_G == 0, G, wall_bc, J=derivative(F_G, G))

G = enforce_lower_bound(G, G_min)
y = 1.0 / G - 1.0 / G0
```

Listing 3.1: Pseudocode for the relaxed Eikonal wall-distance solve.

## 3.4 Variational Formulation and FEniCS Implementation

The standard Galerkin weak formulation is constructed by multiplying the steady transport Equation 3.7 by a test function $\xi$ and applying integration by parts to the diffusive terms. Because the equation is highly non-linear, robust convergence is dependent on the stability of the external velocity field provided by the Picard outer loop.

To manage this non-linearity, the equation is linearized within the Picard iteration scheme by freezing certain parameters using the most recent value of the working variable, $\tilde{\nu}^0$. Specifically, the turbulence production and cross-diffusion terms are treated as explicit sources evaluated using $\tilde{\nu}^0$. Conversely, the wall destruction term is linearized into an implicit reaction sink by scaling the continuous $\tilde{\nu}$ against the frozen variable, taking the form $c_{w1} f_w(\tilde{\nu}^0/y^2)\tilde{\nu}$.

The finite-element implementation mirrors the analytical weak form using the most recent convective velocity field passed from the flow solver. Listing 3.2 shows the residual assembled in each Picard step. The convection, implicit diffusion, and implicit destruction terms are placed on the left-hand side, while production and cross-diffusion are evaluated from the previous SA field and treated as explicit sources.

```
# Unknown nu_tilde, test function xi, frozen previous field nu_tilde0.
S_tilde0 = modified_vorticity(u, nu_tilde0, y)
f_w0 = wall_destruction_function(nu_tilde0, S_tilde0, y)

reaction = c_w1 * f_w0 * nu_tilde0 / y_safe**2
source = (
    c_b1 * S_tilde0 * nu_tilde0
    + (c_b2 / sigma) * inner(grad(nu_tilde0), grad(nu_tilde0))
)

```

```
    F_SA = (
        dot(u, grad(nu_tilde)) * xi * dx
        + inner((nu + nu_tilde0) / sigma * grad(nu_tilde), grad(xi)) * dx
        + reaction * nu_tilde * xi * dx
        - source * xi * dx
    )

    solve(lhs(F_SA) == rhs(F_SA), nu_tilde, sa_boundary_conditions)
    nu_t = eddy_viscosity_from(nu_tilde)
```

Listing 3.2: Pseudocode for the Picard-linearized SA weak-form assembly.

## 3.5 Conclusion

This chapter has established the numerical formulation of the FEniCS-SA flow solver used throughout the remainder of the thesis. The key ingredients are the weak-form finite-element implementation, Picard decoupling of the velocity-pressure and turbulence fields, IPCS pseudo-time stepping for the incompressible flow solve, the steady SA transport equation, and the relaxed Eikonal wall-distance formulation.

Separating these methods from their verification keeps the thesis logic clean: this chapter explains how the solver is constructed, Chapter 4 extends the same finite-element machinery to density-based topology optimization and adjoint sensitivities, and Chapter 5 then verifies the standalone SA solver against Ansys Fluent benchmark cases before the topology optimization results are interpreted.

# Chapter 4

# Turbulent Topology Optimization Methodology

## 4.1 Introduction

Building upon the Spalart-Allmaras (SA) fluid dynamics solver [35, 36] developed in Chapter 3, this chapter introduces the mathematical framework required to perform Topology Optimization (TO) in the turbulent regime. Chapter 3 explained how the forward turbulent flow problem is solved; the present chapter explains how that forward solver is embedded in a density-based design loop and differentiated for gradient-based optimization. The forward solver is verified separately in Chapter 5. The core objective of this methodology is to algorithmically determine the optimal distribution of fluid channels within a given design domain to reduce the pressure loss required to drive a prescribed flow rate.

This chapter details the density-based fictitious-domain formulation used to represent solid and fluid regions on a fixed computational mesh [13, 20]. It then explains the specific turbulent-flow ingredients retained for the remainder of the thesis: a Brinkman-type momentum resistance, a power-law topology penalty for the SA and wall-distance equations, and the reciprocal penalized wall-distance formulation used by the pipe-bend and U-bend computations.

To solve this optimization problem in a computationally tractable manner, continuous adjoint sensitivity analysis is required [31, 30]. The optimization results reported in this thesis use the frozen-turbulence, or Constant Eddy Viscosity (CEV), assumption [46, 34]. This assumption is stated explicitly because it is both the practical reason the turbulent optimization loop remains stable and the main limitation of the sensitivity model.

Following the derivation of these sensitivities, the continuous optimization problem is formally defined, establishing the available loss objectives and the strict fluid volume constraints over explicitly defined design regions. Finally, the mathematical techniques for active-domain density filtering [27] and Heaviside projection [44] are detailed to ensure the generation of manufacturable designs.

## 4.2 The Fictitious Domain Method and Brinkman Penalization

In standard shape optimization, the solid-fluid boundary is explicitly defined by the mesh, and the nodes are physically moved to improve the design. In density-based topology optimization, however, the computational domain ($\Omega$) remains strictly fixed. To create solid walls and fluid channels within this fixed grid, the Fictitious Domain Method is employed.

Originally applied to Stokes flow by Borrvall and Petersson [13] and extended to the Navier-Stokes equations by Gersborg-Hansen et al. [20], this method assigns a continuous design variable to the spatial coordinates in the domain. In the convention used here, a physical topology value of $\bar{\gamma} = 1$ represents a pure **fluid** channel, while $\bar{\gamma} = 0$ represents a **solid**, impermeable structure. To force the fluid velocity to zero inside the solid regions without mathematically removing the finite elements, the fluid is treated as a fictitious porous medium.

This flow resistance is quantified by an inverse permeability term, or Brinkman penalty function, denoted as $\alpha(\bar{\gamma})$. To ensure the optimization algorithm (which relies on continuous gradients) can transition smoothly between fluid and solid states, $\alpha$ must be a continuous interpolation function. The formulation uses a convex Rational Approximation of Material Properties (RAMP) interpolation [37] scaled between a maximum solid penalty ($\alpha_{solid}$) and a minimum fluid penalty ($\alpha_{fluid}$):

$$\alpha(\bar{\gamma}) = \alpha_{solid} + (\alpha_{fluid} - \alpha_{solid})\frac{\bar{\gamma}(1+q)}{\bar{\gamma}+q} \tag{4.1}$$

where $\alpha_{solid}$ is a sufficiently large penalty factor that completely stagnates the flow when $\bar{\gamma} = 0$, and $q$ is a tuning parameter that controls the convexity of the penalization curve. During continuation, $q$ is increased to sharpen the fluid-solid interpolation while keeping the underlying RAMP form differentiable. Unlike traditional structural optimization which relies on the SIMP power-law [12], fluid optimization generally avoids SIMP for momentum because its derivative vanishes at $\bar{\gamma} = 0$. By utilizing the RAMP formulation, a strictly non-zero gradient is guaranteed across the entire design domain.

### 4.2.1 The Density Chain and Domain Bounding

While conceptually represented by the physical topology field in the governing equations, the layout is managed through a distinct chain of density representations to handle design and non-design domains. To explicitly avoid mathematical confusion with the fluid mass density ($\rho$) utilized in the Reynolds-Averaged Navier-Stokes equations, this thesis exclusively utilizes the $\gamma$ notation for all topological variables:

- **Raw design density** ($\gamma$): The baseline parameter manipulated by the optimizer.
- **Filtered density** ($\tilde{\gamma}$): The spatially smoothed design field.

- **Projected effective density** ($\bar{\gamma}$): The bounded, physical topology field evaluated by the governing equations.

The raw design density, $\gamma$, is defined as a Discontinuous Galerkin (DG0) [23] function over the entire mesh. The optimization algorithm strictly manipulates the degrees of freedom corresponding to the actively optimized design space, while passive non-design cells are subsequently restored to their prescribed bounding values.

This raw variable is subjected to a spatial PDE filter [27] to yield a smoothed field $\tilde{\gamma}$. To recover sharp boundaries and strictly preserve required non-design regions, the filtered density is then projected and mapped into the bounded effective density ($\bar{\gamma}$) [22]:

$$\bar{\gamma} = \gamma_{\text{lower}} + (\gamma_{\text{upper}} - \gamma_{\text{lower}})\Pi_{\beta}(\tilde{\gamma}) \tag{4.2}$$

where $\Pi_{\beta}$ is the projection operator, and $\gamma_{\text{lower}}$ and $\gamma_{\text{upper}}$ define the absolute permissible bounds for any given cell. This rigorous bounding guarantees that predefined features, such as non-design fluid inlet extensions ($\gamma_{\text{lower}} = \gamma_{\text{upper}} = 1$) or explicitly constrained structural walls ($\gamma_{\text{lower}} = \gamma_{\text{upper}} = 0$), remain rigidly fixed without interfering with the mathematical behavior of the active optimization space.

The computational mesh is therefore partitioned into an active design domain and passive non-design domains. The active design domain contains the cells whose raw densities are updated by MMA. Fluid non-design domains are fixed as permanent fluid, typically to preserve inlet and outlet extensions, buffer channels, or measurement regions. Solid non-design domains are fixed as permanent solid, typically to impose external walls, blocked regions, or geometric features that must not be modified. These passive cells still participate in the forward flow, turbulence, and wall-distance solves, but they are excluded from the optimization masks: the volume constraint is evaluated only over the prescribed volume mask $\Omega_V$, and any domain-integrated post-processing metric is evaluated only over its prescribed integration mask. Consequently, prescribed inlet/outlet buffers and mandatory solid walls do not count toward the optimized fluid fraction and do not directly bias the reported domain-integrated metrics.

## 4.3 Penalization of the Governing Equations

To effectively optimize turbulent manifolds, the respective penalties must be rigorously integrated into the turbulence model and the wall-distance calculations to ensure the physics remain consistent as the solid boundaries evolve. All physical penalties strictly utilize the projected effective density, $\bar{\gamma}$.

### 4.3.1 Modified Reynolds-Averaged Navier-Stokes Equations

The primary application of the Brinkman penalty is the modification of the steady-state, incompressible RANS momentum equation. The RAMP-interpolated inverse permeability $\alpha(\bar{\gamma})$ is added as a linear sink term (Darcy friction) to the momentum conservation equation:

$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nabla \cdot \left[(\mu + \mu_t)\left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T\right)\right] - \alpha(\bar{\gamma})\mathbf{u} \tag{4.3}$$

It is critical to note the retention of the full symmetric rate-of-strain tensor. Here, $\rho$ strictly denotes the fluid mass density. Because the turbulent eddy viscosity ($\mu_t$) varies drastically across the spatial domain [32], the effective fluid viscosity is no longer constant, and the transposed term exerts a non-zero physical influence on the momentum balance. In regions where the optimizer places solid material ($\bar{\gamma} \to 0$), the enormous magnitude of $\alpha_{solid}\mathbf{u}$ dominates all inertial and viscous forces, forcing the local velocity mathematically to zero.

### 4.3.2 Modified Spalart-Allmaras Transport Equation

As the Fictitious Domain Method forces velocity to zero in the solid regions, the turbulence generation inside those regions should theoretically cease. However, due to numerical diffusion, turbulent eddy viscosity ($\tilde{\nu}$) can inadvertently bleed into the solid regions. This unphysical behavior destabilizes the solver and heavily distorts the adjoint sensitivity analysis.

To strictly enforce a laminar, zero-turbulence state inside the solid structures, a corresponding penalty sink term is added to the Spalart-Allmaras transport equation [47, 16]:

$$\begin{aligned}\mathbf{u} \cdot \nabla\tilde{\nu} = c_{b1}\tilde{S}\tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{y}\right)^2 + \frac{1}{\sigma}\left[\nabla \cdot ((\nu + \tilde{\nu})\nabla\tilde{\nu}) + c_{b2}(\nabla\tilde{\nu})^2\right] \\ - \alpha_{\tilde{\nu}}(1-\bar{\gamma})^{N_{\tilde{\nu}}}\tilde{\nu}\end{aligned} \tag{4.4}$$

The last term is the power-type topology penalty used to suppress the SA working variable in solid material. Because $\bar{\gamma} = 1$ denotes fluid and $\bar{\gamma} = 0$ denotes solid, the penalty is applied to the local solid fraction $(1 - \bar{\gamma})$. The coefficient $\alpha_{\tilde{\nu}}$ sets the damping strength and the exponent $N_{\tilde{\nu}}$ controls how sharply the damping grows as a cell becomes solid. In the final benchmark computations these two parameters are held constant throughout the optimization. The term is therefore zero in pure fluid, reaches its full strength in solid cells, and acts only as an artificial damping device for the working variable, not as an additional turbulence-model closure.

### 4.3.3 Reciprocal Penalized Wall-Distance Equation

The Spalart-Allmaras model is highly sensitive to the wall distance, $y$. While differential equations for wall distance computation are well-established [42], topology optimization introduces a unique challenge. The location of the wall is not a static mesh boundary, but rather the dynamically evolving $\bar{\gamma} = 0$ contour. Therefore, the relaxed Eikonal equation must be penalized to dynamically recognize the fictitious solid as a physical boundary [47]:

$$\nabla G \cdot \nabla G + \sigma_w G(\nabla \cdot \nabla G) - (1 + 2\sigma_w)G^4 - \alpha_G(1-\bar{\gamma})^{N_G}(G - G_0) = 0 \tag{4.5}$$

The wall-distance equation uses the same power-law topology penalty. The coefficient $\alpha_G$ sets the strength of the artificial wall-distance forcing and the exponent $N_G$ controls its growth through gray material. As for the SA penalty, these parameters are held constant in the final benchmark computations. In fluid regions the term vanishes, while in topology-created solid regions it pulls $G$ toward $G_0$. The algebraic recovery function

$$y = \frac{1}{G} - \frac{1}{G_0} \tag{4.6}$$

then collapses the SA wall distance toward zero inside topology-created solid regions while retaining the physical wall distance in fluid regions.

## 4.4 Boundary Conditions and Turbulence Initialization

To stabilize the topology optimization and ensure physical accuracy, boundary constraints are strictly managed across both fluid and turbulence regimes. Non-design fluid buffer regions are strictly mandated at inlets and outlets to isolate the active optimization domain from artificial numerical shear. The framework robustly handles boundary interactions by incorporating pressure outlets coupled with optional velocity-component constraints, as well as pressure pin controls to fully constrain the discrete PDE problem.

At the domain inlets, establishing accurate turbulent boundary conditions natively bridges physical intent with numerical stability. The prescribed turbulence intensity and the associated turbulence length scale are systematically converted into an intermediate eddy-viscosity ratio. This ratio directly calculates the requisite boundary value for the Spalart-Allmaras working variable $\tilde{\nu}$. Throughout the computation, to protect the solver from chaotic geometric fluctuations during intermediate design cycles, rigorous numerical floors, ceilings, and internal clipping functions are imposed to bind the working variable safely within physical limits.

## 4.5 Sensitivity Analysis and the Frozen Turbulence Assumption

With the forward state equations fully defined and penalized, the topology optimization algorithm requires the gradient (sensitivity) of the objective function with respect to the design variable. Because evaluating finite differences for every mesh element is computationally impossible, the continuous adjoint method is employed [30].

However, calculating the fully coupled, exact adjoint for turbulent flows is mathematically formidable. As demonstrated by Zymaris et al. [51], formulating the exact continuous adjoint for the Spalart-Allmaras model requires differentiating the highly non-linear production and destruction terms, accounting for the variations in the distance function, and managing the intense cross-coupling with the momentum equations. The complexity of these exact adjoints frequently introduces severe numerical stiffness and divergence.

To make the gradient calculation both numerically robust and computationally feasible, the primary optimization workflows rely on the **Frozen Turbulence As-**

**sumption**, frequently referred to in the literature as the Constant Eddy Viscosity (CEV) assumption [46, 34].

### 4.5.1 Mechanism of Frozen Turbulence

The frozen turbulence assumption simplifies the sensitivity analysis by decoupling the turbulence transport from the design update. During the adjoint formulation, the turbulent eddy viscosity field ($\nu_t$) generated by the forward SA solver is assumed to be locally independent of infinitesimally small changes in the physical topology ($\bar{\gamma}$). Mathematically, this implies that during the calculation of the adjoint sensitivities:

$$\frac{\partial \nu_t}{\partial \bar{\gamma}} \approx 0 \tag{4.7}$$

By treating the eddy viscosity field as frozen (a spatially varying but constant parameter), the complex adjoints of the Spalart-Allmaras and Eikonal equations do not need to be solved. The optimizer only solves the standard fluid adjoint equations (utilizing the effective viscosity $\mu + \mu_t$) to determine how a change in topology affects the selected pressure-loss objective.

### 4.5.2 Justification and Limitations

The primary advantage of the frozen turbulence approach is not that it is universally better, but that it makes the optimization problem solvable within a reasonable numerical and implementation budget. Relative to a fully coupled turbulent adjoint, it offers greater numerical stability, a smaller implementation burden, and a materially lower computational cost per design iteration. This is why frozen turbulence remains common in industrial adjoint workflows and why it is used as the baseline in this thesis [30, 34].

The price of this robustness is that the gradients are approximate reduced derivatives. Because the algorithm ignores how an infinitesimal topology change would alter future turbulence production, the sensitivity can be inaccurate when the objective is strongly controlled by separated shear layers, stall-like behavior, or other regimes where the eddy viscosity changes sharply with geometry. Wu et al. [46] show that the CEV assumption can still achieve useful optimization goals in benign flows, but its sensitivity accuracy and adjoint convergence can deteriorate near stall and in unsteady settings. Zymaris et al. [51] further demonstrate, for SA-based continuous adjoints, that including the turbulence-equation adjoint improves derivative accuracy for loss functionals. The present thesis therefore defends the frozen assumption as a practical and verifiable approximation, not as an exact turbulence sensitivity model. Attempts to include the SA turbulence equation in the adjoint were explored during the implementation phase, but the resulting solves were not sufficiently stable for the thesis optimization campaign.

## 4.6 Optimization Problem Formulation

Using the frozen-turbulence sensitivities described above, the topology optimization task is formulated as a mathematical programming problem, where the optimizer must find a spatial distribution that minimizes a specific performance metric subject to physical and geometric constraints.

### 4.6.1 Objective Formulation

Dissipation-type objectives are common in density-based flow topology optimization because they penalize mechanical energy loss directly. In Brinkman-penalized formulations, the corresponding power-loss objective is naturally written as the sum of physical viscous dissipation and Darcy/Brinkman drag over the optimized design domain [47, 16]:

$$J_D(\mathbf{u},\bar{\gamma}) = \int_{\Omega_D} \left[\frac{1}{2}\mu_{\text{eff}}\left(\nabla\mathbf{u}+\nabla\mathbf{u}^T\right) : \left(\nabla\mathbf{u}+\nabla\mathbf{u}^T\right) + \alpha(\bar{\gamma})\,\mathbf{u}\cdot\mathbf{u}\right] d\Omega. \tag{4.8}$$

Here $\Omega_D$ denotes the optimized design domain used for the objective integration, excluding the fixed non-design inlet and outlet extensions, and $\mu_{\text{eff}} = \mu + \mu_t$ under the frozen-turbulence flow solve. The first term measures viscous deformation losses in the design region, while the Brinkman term penalizes residual velocity in low-density regions where $\alpha(\bar{\gamma})$ is large. Unlike a pure post-processing metric, $J_D$ therefore contains an explicit local design-dependent contribution through $\alpha(\bar{\gamma})$.

For the pipe-bend and U-bend benchmarks considered in this thesis, the final engineering goal is instead expressed as a global pressure-loss objective: guide a prescribed inflow from inlet to outlet with the lowest possible hydraulic resistance. Because the outlet pressure is fixed to zero, this loss is represented directly by the average pressure on the physical inlet boundary, where the inlet velocity is prescribed,

$$J_p(p) = \frac{1}{|\Gamma_{\text{in}}|}\int_{\Gamma_{\text{in}}} p\, d\Gamma. \tag{4.9}$$

Here $\Gamma_{\text{in}}$ is the inlet boundary of the fixed non-design extension, not an internal design-domain pressure plane. When the outlet pressure is used as the gauge reference, minimizing $J_p$ is equivalent to minimizing the pressure loss required to drive the imposed flow rate through the topology. This is a boundary integral rather than a volume integral over $\Omega_D$, and it does not include an explicit Brinkman-drag term in the objective. The Brinkman penalization still enters the optimization through the state equation in Eq. 4.3; changing $\bar{\gamma}$ changes the resistance field, the forward pressure solution, and the adjoint sensitivity. The pressure objective is therefore less explicitly penalized than $J_D$, but it remains a valid hydraulic-loss objective for fixed-flow problems. Its sensitivity is mapped back only to the active design degrees of freedom through the same active-mask filter and transpose-filter machinery, so the prescribed non-design inlet extension is used only to impose the velocity and measure the pressure and is not itself optimized. Physical viscous dissipation is retained as both an alternative objective definition and a validation metric in Chapter 6; the final benchmark runs use the pressure objective $J_p$.

### 4.6.2 Volume Constraint

Because the goal is to design a targeted manifold or pipe system, a strict volume constraint must be imposed. The volume fraction constraint is evaluated only over the prescribed volume mask $\Omega_V$, rather than over the entire computational mesh:

$$g(\bar{\gamma}) = \frac{\int_{\Omega_V} \bar{\gamma}\, d\Omega}{V_{\text{mask}}} - V_{\max} \leq 0 \tag{4.10}$$

where $V_{\text{mask}}$ is the total volume of the masked region and $V_{\max}$ is the maximum allowable fluid volume fraction inside that region. Fixed non-design fluid domains, such as inlet and outlet extensions, are therefore not included in either the numerator or denominator of the volume constraint. They remain fluid for the flow solve, but they do not count as optimized fluid volume and cannot make the active design region satisfy the volume constraint artificially.

### 4.6.3 The Complete Continuous Optimization Problem

Combining the selected objective function, the governing equations, and the constraints, the continuous topology optimization problem for turbulent flow is formally stated as:

$$\begin{aligned}
\underset{\gamma(\mathbf{x})\in[0,1]}{\text{minimize}} \quad & J(\mathbf{u}, p, \bar{\gamma}) \\
\text{subject to} \quad & \text{Modified RANS Equations (Eq. 4.3)} \\
& \text{Modified Spalart-Allmaras Equation (Eq. 4.4)} \\
& \text{Penalized Wall-Distance Equation (Eq. 4.5)} \\
& g(\bar{\gamma}) \leq 0
\end{aligned} \tag{4.11}$$

where $J$ denotes the selected loss objective, either the dissipation objective $J_D$ in Eq. 4.8 or the inlet-pressure objective $J_p$ in Eq. 4.9. The wall-distance constraint used for the final benchmark computations is the reciprocal penalized relaxed Eikonal equation in Eq. 4.5, with the wall distance recovered by Eq. 4.6.

To solve this constrained minimization problem, the Method of Moving Asymptotes (MMA) [39] is used. MMA is specifically designed for handling large-scale, gradient-based topology optimization problems. It approximates the highly nonlinear design space using convex asymptotes, relying on strict move limits which cap the maximum allowable change in the active raw design variable ($\gamma$) during any single optimization step.

## 4.7 Density Filtering and Projection

A well-known mathematical complication in density-based topology optimization is the lack of existence of a classical solution. If the design space is left unconstrained, the optimizer will tend to create infinitely thin micro-structures (checkerboarding). To guarantee a mesh-independent and manufacturable design, a minimum length scale must be enforced.

### 4.7.1 The Discontinuous Galerkin Helmholtz Filter

To prevent checkerboarding, a density filter is applied to the raw design variable generated by the optimizer. Because a continuous nodal definition causes the Brinkman penalty to interpolate smoothly across the interior of boundary elements, creating a semi-permeable gray gradient that allows mass leakage, the density field is discretized using piecewise constant elements (Discontinuous Galerkin, DG0) [9]. By assigning a single, uniform density value to the entire element, the structural boundaries remain perfectly sharp at the element edges.

Consequently, the standard continuous Helmholtz filter cannot be applied. The filter is mathematically adapted to account for the discontinuous boundaries utilizing an interior penalty method across the element facets ($\Gamma_{int}$). Standard volume diffusion is mathematically replaced by interior facet jump penalties:

$$\int_{\Gamma_{int}} r^2 \frac{\alpha_{dg}}{h_{avg}} [\![\tilde{\gamma}]\!] \cdot [\![v]\!] \, dS + \int_{\Omega} \tilde{\gamma} v \, dx = \int_{\Omega} \gamma v \, dx \tag{4.12}$$

where $v$ is the test function, $[\![\cdot]\!]$ denotes the jump operator across a facet, $h_{avg}$ is the average cell size, and $r = \frac{R}{2\sqrt{3}}$ is the scaled filter radius.

#### Active-Mask Normalization

To appropriately handle the severe boundary interactions near designated inlet and outlet buffers, this filtering is strictly active-mask-normalized. Rather than applying the filter blindly across the entire domain, the filtered output evaluates as:

$$\tilde{\gamma} = \frac{H(M\gamma)}{H(M)} \tag{4.13}$$

where $H$ represents the Helmholtz PDE operator and $M$ represents the isolated active design indicator mask. Crucially, during sensitivity analysis, the mathematical transpose of this normalized filter must be explicitly applied to map objective gradients back into the active DG0 design space. Listing 4.1 summarizes the forward and transpose operations without reproducing implementation details.

```
    # Forward filter: only active cells influence the smoothed field.
    denominator = helmholtz_solve(active_mask)
    numerator = helmholtz_solve(active_mask * gamma_raw)
    gamma_filtered = safe_divide(numerator, denominator)

    # Projection and hard restoration of prescribed non-design cells.
    gamma_projected = heaviside_projection(gamma_filtered, beta, eta)
    gamma_effective = gamma_lower + (gamma_upper - gamma_lower) *
gamma_projected
    gamma_effective[passive_fluid_cells] = 1.0
```

```
        gamma_effective[passive_solid_cells] = 0.0

        # Transpose filter: map adjoint sensitivities back to active cells.
        dJ_dgamma_projected = dJ_dgamma_effective * (gamma_upper - gamma_lower)
        dJ_dgamma_filtered = dJ_dgamma_projected * projection_slope(gamma_filtered
    , beta, eta)
        filter_rhs = active_mask * safe_divide(dJ_dgamma_filtered, denominator)
        dJ_dgamma_raw = helmholtz_solve(filter_rhs)
        dJ_dgamma_raw[inactive_cells] = 0.0
```

Listing 4.1: Pseudocode for active-mask-normalized density filtering and transpose-gradient filtering.

### 4.7.2 Heaviside Projection

While the Helmholtz filter successfully imposes a minimum length scale, it inherently smooths the design field, resulting in a large gray transition zone between the solid and fluid regions. To recover a crisp, discrete topology, a continuous Heaviside projection is applied to the filtered variable [44, 22]:

$$\Pi_\beta(\tilde{\gamma}) = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{\gamma} - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))} \tag{4.14}$$

where $\eta \in [0, 1]$ is the projection threshold (typically 0.5), and $\beta$ is a steepness parameter. As $\beta \to \infty$, the function approaches a strict Heaviside step function.

## 4.8 Algorithmic Workflow and Solver Implementation

Executing the optimization requires a highly robust discrete, iterative algorithm. To support the severe non-linearities of the turbulent regime, the solver architecture natively manages the segregated Spalart-Allmaras turbulence model, explicit physics decoupling via Picard iterations, and deeply hybridized forward flow routines.

### 4.8.1 Boundary Conditions and Turbulence Initialization

At the domain inlets, establishing accurate turbulent boundary conditions natively bridges physical intent with numerical stability. The prescribed turbulence intensity and the associated turbulence length scale are systematically converted into an intermediate eddy-viscosity ratio. This ratio directly calculates the requisite boundary value for the Spalart-Allmaras working variable $\tilde{\nu}$. Throughout the computation, to protect the solver from chaotic geometric fluctuations during intermediate design cycles, rigorous numerical floors, ceilings, and internal clipping functions are imposed to bind the working variable safely within physical limits.

Furthermore, the solver dynamically handles pressure outlets coupled with optional velocity-component constraints, allowing the flow field to adapt to the evolving topology without artificial boundary blockages.

### 4.8.2 SA Numerical Solution Controls

The SA transport solve is controlled through Picard-level coupling, carefully bounded turbulence variables, continuation of the topology penalties, and the hybrid forward-flow recovery strategy described below.

### 4.8.3 Hybrid Forward Flow Recovery and Picard Decoupling

Rather than relying solely on simple fixed-point decoupled sweeps, the forward fluid solver utilizes an advanced hybrid architecture integrating both Incremental Pressure Correction Scheme (IPCS) methods and monolithic SNES workflows.

During the optimization loop, a Picard iteration scheme strictly decouples the fluid and turbulence physics. The flow solver advances the momentum field using the eddy viscosity from the previous iteration, and this updated velocity is then passed back to the SA solver. During volatile intermediate design steps, the fluid framework initiates convection continuation strategies and leverages Stokes-Brinkman warm starts to slowly reintroduce inertial physics. If convergence stalls, the SNES solver autonomously steps through multiple recovery schedules (including dynamic tolerance reductions and altered line-search methods).

The optimization architecture supports non-converged acceptance, allowing the algorithm to passively absorb intermediate errors and maintain topological momentum. However, prior to the continuous adjoint assembly, stricter flow solves can be executed. When used, these strict solves confirm that the sensitivities, and specifically the frozen turbulent fields, are mathematically derived from a fully equilibrated state. Otherwise, relying on the accepted residual criteria serves as a necessary pragmatic compromise to maintain solver momentum in highly separated regimes.

### 4.8.4 Algorithm and Continuation Strategy

To ensure convergence to a physically meaningful binary design, the algorithm iterates over a strict continuation schedule. As the optimization progresses, the Brinkman penalty parameter ($q$) and the projection steepness ($\beta$) are iteratively increased, while the MMA move limits are proportionally tightened. The SA and wall-distance power penalties do not use this continuation in the final benchmark runs; their amplitudes and exponents remain fixed while the local penalty magnitude changes only through the evolving topology field.

The global optimization routine is outlined in Listing 4.2. The listing keeps the algorithmic sequence visible: continuation, density processing, wall-distance update, Picard-decoupled turbulent flow solve, frozen-turbulence adjoint assembly, transpose filtering, and MMA update.

```
for stage in continuation_schedule:
    ramp_convexity = stage.ramp_convexity
    beta = stage.projection_steepness
    move_limit = stage.move_limit
    set_fixed_sa_and_wall_penalties()

```

```
        while not mma_converged and inner_iter < stage.max_iterations:
            gamma_filtered = density_filter(gamma_raw, active_mask)
            gamma_effective = project_and_restore_bounds(gamma_filtered, beta)
            alpha = brinkman_ramp(gamma_effective, solid_resistance,
                                  fluid_resistance, ramp_convexity)

            wall_distance = update_wall_distance(
                gamma_effective,
                wall_penalty_strength,
                wall_penalty_exponent,
            )

            for picard_iter in range(max_picard_iterations):
                velocity, pressure = solve_flow(alpha, nu_t)
                nu_tilde = solve_spalart_allmaras(
                    velocity,
                    wall_distance,
                    sa_penalty_strength,
                    sa_penalty_exponent,
                )
                nu_t = eddy_viscosity_from(nu_tilde)
                if residuals_are_accepted():
                    break

            nu_t_frozen = freeze_eddy_viscosity(nu_t)
            velocity, pressure = solve_final_state(alpha, nu_t_frozen)
            adjoint = solve_frozen_turbulence_adjoint(velocity, pressure, nu_t_frozen)
            dJ_dgamma_effective = assemble_design_sensitivity(adjoint, gamma_effective)
            dJ_dgamma_raw = transpose_filter(dJ_dgamma_effective, active_mask)
            gamma_raw[active_design_cells] = MMA.update(
                gamma_raw[active_design_cells], dJ_dgamma_raw, move_limit
            )
```

Listing 4.2: Pseudocode for the frozen-turbulence topology optimization loop.

## 4.9 Conclusion

This chapter has established the mathematical and algorithmic framework used for the turbulent topology optimization results in this thesis. By employing the Fictitious Domain Method with Brinkman momentum resistance [13, 20], the evolving structural topology is integrated into the fixed computational grid. The SA working-variable and reciprocal wall-distance equations are coupled to the topology through the power-law penalty, while the DG0 Helmholtz interior penalty filter [27] and continuous Heaviside projection [44] enforce a minimum length scale and reduce mesh-dependent numerical artifacts.

The optimization is driven by the frozen-turbulence continuous adjoint [46, 34]. This assumption is defensible as a stable engineering approximation, but the thesis explicitly treats it as an approximation whose limitations must be checked through finite-difference sensitivity verification and independent CFD validation. Embedded within a Picard-decoupled flow architecture using hybrid IPCS/SNES recovery schemes, this methodology transforms the fluid dynamics engine from Chapter 3 into the automated design tool whose forward behavior is verified in Chapter 5 and whose optimized designs are evaluated in Chapter 6.

# Chapter 5

# Verification of the Spalart-Allmaras Solver for Turbulent Internal Flows

## 5.1 Introduction

Chapter 3 introduced the numerical implementation of the Spalart-Allmaras (SA) model in FEniCS. Before using this solver inside the topology optimization framework, the forward fluid solver is verified against independent Ansys Fluent calculations on two benchmark flows. The first benchmark is a fully developed turbulent channel, which isolates wall damping, turbulence production, and Picard coupling in a simple geometry. The second benchmark is a curved U-bend, which tests the same implementation under strong convection, streamline curvature, and separation.

This chapter focuses on the verification evidence rather than the full simulation setup. For the channel benchmark, the physical setup is listed in Table A.1, the FEniCS solver settings in Table A.2, and the mesh studies in Tables A.3 and A.4 of Section A.1. For the U-bend benchmark, the corresponding geometry, numerical settings, truncation study, and mesh studies are documented in Tables A.5, A.6, A.7, and A.8, together with Figure A.3 in Section A.2. The purpose here is to show that the FEniCS-SA implementation reproduces the relevant Ansys-SA flow behavior well enough to serve as the forward model for the topology optimization studies in Chapter 6.

## 5.2 Verification Benchmark 1: Channel Flow

To verify the fundamental correctness of the custom fluid and turbulence solvers, the code was first tested on a standard 2D turbulent channel flow. To cleanly isolate and verify turbulence production, wall damping, and Picard coupling without the artifacts of a developing boundary layer, this benchmark was formulated as a pressure-driven periodic channel rather than a conventional inlet-outlet duct.

The velocity field ($\mathbf{u}$) and Spalart-Allmaras working variable ($\tilde{\nu}$) were constrained periodically between the streamwise boundaries, while the top and bottom boundaries were treated as standard no-slip walls ($\mathbf{u} = 0, \tilde{\nu} = 0$). Flow was driven entirely by a fixed pressure drop from the upstream plane ($p = 2$ Pa) to the downstream plane ($p = 0$ Pa). Based on a reference bulk velocity of $U_{\text{ref}} = 18.5$ m/s and using the full channel height ($H = 2.0$ m) as the characteristic length scale, the solver was targeted to resolve a fully turbulent flow regime at a channel Reynolds number of $Re_H \approx 20{,}300$. The complete definition of the geometric, physical, and boundary-condition parameters is provided in Table A.1 of Section A.1.1 in Appendix A.

### 5.2.1 Mesh Resolution Requirements

Because the SA formulation is a low-Reynolds model that does not use empirical wall functions, it requires mathematical integration directly through the viscous sublayer. To satisfy this strict boundary-layer requirement, a custom structured inflation mesh was generated for the channel flow. Because the domain is a simple periodic rectangle, this anisotropic grid was constructed programmatically using NumPy arrays and converted into a FEniCS-compatible format via the `meshio` library. Based on the target bulk Reynolds number, this approach enforced a physical first-cell height of approximately $1.7 \times 10^{-3}$ m, aiming for a non-dimensional wall distance of $y^+ \approx 1$ across the entire domain.

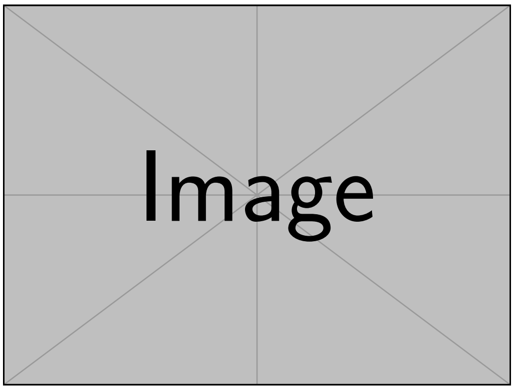
To verify that the numerical results were not dominated by spatial discretization, a grid-independence study was conducted. The full spatial verification is documented in Section A.1.3 of Appendix A. The mesh hierarchy is detailed in Table A.3, and visual confirmation of convergence is provided by the velocity profiles in Figure A.1.

### 5.2.2 FEniCS vs. Ansys Comparison

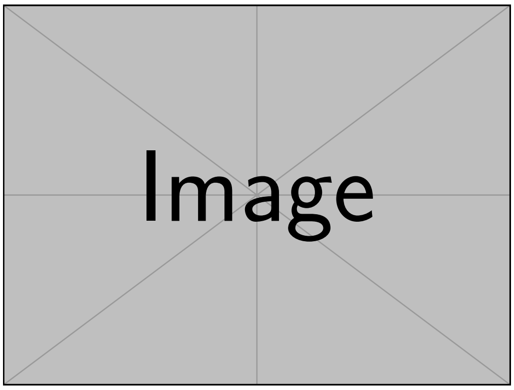
Prior to this comparative analysis, the Ansys baseline itself was verified for grid independence to ensure a robust reference (see Table A.4 and Figure A.2 in Section A.1.3 of Appendix A). The channel flow was then simulated in FEniCS using the decoupled Picard iteration method and compared against the Ansys Fluent baseline.

As shown in Figure 5.1, the qualitative velocity contours of the custom FEniCS implementation show good agreement with the Ansys Fluent baseline. The quantitative line plots confirm that the FEniCS velocity profiles closely track the Ansys reference data. The custom solver resolves the logarithmic region of the turbulent boundary layer and predicts the turbulent diffusion toward the channel center. The quantitative errors and macroscopic metrics are summarized in Table 5.1.

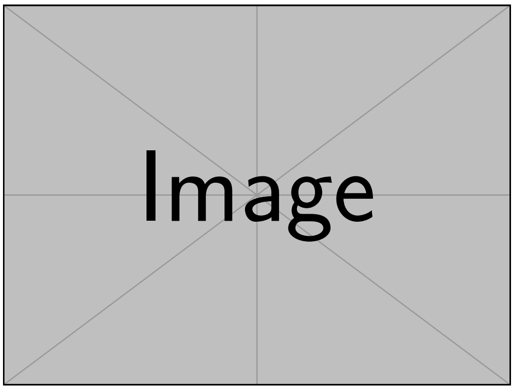
This agreement supports the verification that the core Navier-Stokes solver, relaxed Eikonal wall-distance field, and SA production/destruction terms behave correctly in a canonical wall-bounded turbulent flow.




(A) Ansys Fluent SA contour




(B) FEniCS SA contour




(C) Velocity profiles

Figure 5.1: Verification of the 2D channel flow ($Re_H \approx$ 20,300). The comparison includes qualitative velocity contours from Ansys Fluent and FEniCS, plus quantitative velocity profiles extracted across the channel.

Table 5.1: Quantitative comparison of flow metrics and extracted errors for the 2D channel benchmark.

| **Metric** | **Ansys Fluent** | **Custom FEniCS** | **Relative Error / Difference** |
|---|---|---|---|
| Bulk Velocity ($U_{\text{bulk}}$) [m/s] | [TBD] | [TBD] | [TBD]% |
| Friction Coefficient ($C_f$) | [TBD] | [TBD] | [TBD]% |
| Max Pointwise Velocity Error | – | – | [TBD]% |
| Relative $L_2$ Velocity Error | – | – | [TBD]% |

## 5.3 Verification Benchmark 2: U-Bend Geometry

Following the channel flow verification, the solver was tested on the highly convective separating U-bend flow evaluated in Chapter 2. This case is closer to the curved internal-flow topologies studied later in the thesis because the flow experiences strong streamline curvature and pressure-gradient-driven redistribution.

As discussed in Section A.2.3 of Appendix A, the verification used a modified computational domain for efficiency. First, the domain was reduced to a 2D geometry corresponding to the symmetry mid-plane of the full 3D case. Since the FEniCS domain is mathematically a 2D planar approximation, it cannot resolve true 3D Dean secondary vortices. Instead, this benchmark validates the 2D symmetry-plane approximation and the solver's ability to capture curvature-induced separation and momentum redistribution. Second, the extended straight inlet and outlet sections of the standard benchmark were truncated to $30D$ to reduce unnecessary computational cost. The mathematical justification for this truncation is plotted in Figure A.3.

To drive the flow, a uniform prescribed velocity ($\mathbf{u}_{in} = 1.42$ m/s) was imposed at the inlet alongside a fully turbulent boundary value ($\tilde{\nu}_{in} = 9.34 \times 10^{-5}$ m$^2$/s), while a zero relative static pressure ($p = 0$ Pa) was assigned to the outlet. While the domain is 2D, the 3D physical pipe diameter ($D = 0.028$ m) and radius of curvature ($R_c = 0.125$ m) were retained as the characteristic length scales. This preserves the target Reynolds number ($Re_D \approx 44{,}700$) and Dean number ($De \approx 15{,}000$) of the 3D Ansys reference case. The physical setup and boundary conditions are listed in Table A.5 of Section A.2.1 in Appendix A.

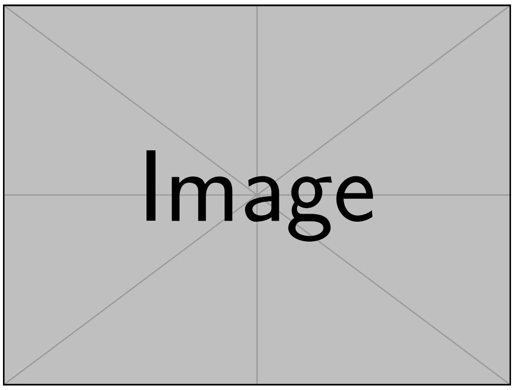
### 5.3.1 Mesh Resolution Requirements

The U-bend geometry features curved inner and outer arcs, complex corner transitions, and anisotropic boundary-layer inflation around non-orthogonal walls. To manage this geometric complexity while satisfying the $y^+ \approx 1$ resolution requirement, a custom mesh was generated using Gmsh [21]. The first element height adjacent to the wall was controlled to $1.20 \times 10^{-5}$ m across the entire domain. The independence studies are documented in Section A.2.4 of Appendix A: the inflation parameters and element densities of the Gmsh hierarchy are listed in Table A.7, and the extracted velocity profiles confirming grid convergence at the bend exit are illustrated in Figure A.4.

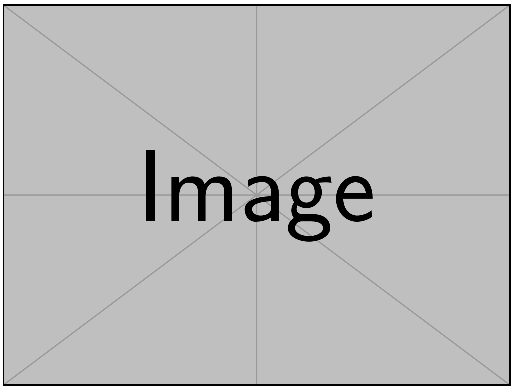
### 5.3.2 FEniCS vs. Ansys Comparison

Having validated the spatial resolution of the FEniCS mesh, an equivalent grid-independence study was executed for the commercial Ansys Fluent baseline (detailed in Table A.8 and Figure A.5 of Section A.2.4 in Appendix A). With both solvers operating on spatially converged grids, an independent Ansys Fluent simulation was executed on the shortened 2D domain.

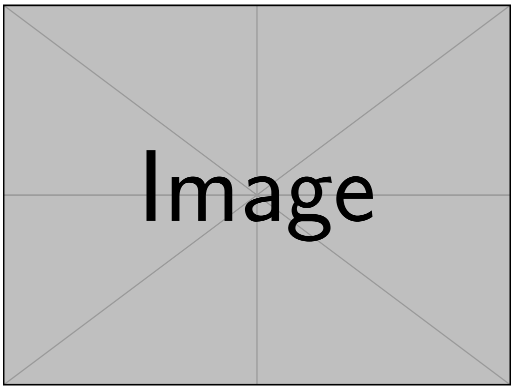
As shown in Figure 5.2, the global flow fields and separation regions observed in the qualitative velocity contours demonstrate visual agreement. To provide a more robust assessment, velocity data was extracted along a cross-sectional line probe positioned at the bend exit. The quantitative velocity profile computed by the custom FEniCS implementation traces the commercial Ansys results closely, adequately capturing the boundary-layer growth and the momentum shift toward the outer wall.




(A) Ansys Fluent SA contour




(B) FEniCS SA contour




(C) Velocity profiles

Figure 5.2: Verification of the 2D U-bend flow ($Re_D \approx 44{,}700$, $De \approx 15{,}000$). The comparison includes qualitative velocity contours from Ansys Fluent and FEniCS, showing separation characteristics, plus quantitative velocity profiles extracted at the bend exit.

Table 5.2 summarizes the quantitative error margins and the macroscopic pressure drop ($\Delta p$) across the 2D U-bend domain for both solvers.

Table 5.2: Quantitative comparison of flow metrics and extracted errors for the 2D U-bend benchmark.

| **Metric** | **Ansys Fluent** | **Custom FEniCS** | **Relative Error / Difference** |
|---|---|---|---|
| Pressure Drop ($\Delta p$) [Pa] | [TBD] | [TBD] | [TBD]% |
| Max Pointwise Velocity Error | – | – | [TBD]% |
| Relative $L_2$ Velocity Error | – | – | [TBD]% |

This general agreement in localized velocity gradients and global pressure-drop behavior indicates that the framework provides a credible forward-flow basis for the pressure-drop minimization objectives in the subsequent topology optimization campaigns.

## 5.4 Conclusion

The two verification cases establish the practical operating range of the custom FEniCS-SA solver before it is embedded in the topology optimization loop. The periodic channel confirms that the solver reproduces canonical wall-bounded turbulent behavior, while the U-bend checks the same implementation in a curved, separation-prone internal flow.

These results do not remove the known limitations of a 2D SA model: secondary Dean vortices, anisotropic Reynolds stresses, and full three-dimensional separation physics remain outside the present solver scope. They do, however, show that the implemented SA equations, wall-distance calculation, and segregated coupling strategy are sufficiently consistent with Ansys Fluent to support the topology optimization methodology and results presented in Chapters 4 and 6.

# Chapter 6

# Turbulent Topology Optimization Results

## 6.1 Introduction

Building upon the FEniCS-SA solver verified in Chapter 5 and the topology optimization methodology developed in Chapter 4, this chapter evaluates the turbulent topology optimization solver on the pipe-bend and U-bend internal-flow benchmarks from Bayat et al. [11]. Both benchmarks use the final thesis formulation: reciprocal wall-distance penalization, topology-dependent SA and wall-distance reaction terms, an average inlet-pressure objective, and the frozen-turbulence adjoint.

The chapter first verifies the implemented frozen sensitivity on the same pipe-bend configuration used for optimization with a Dilgen-style coordinate finite-difference check on selected design cells. The optimized pipe-bend and U-bend layouts are then interpreted through their projected density fields, velocity fields, convergence histories, and independent sharp-geometry CFD validation. This ordering keeps the derivative evidence close to the optimization result that uses it, while leaving the standalone forward-flow verification in Chapter 5.

The U-bend serves as a companion topology optimization benchmark with the same governing assumptions and objective structure, but with a more severe 180° flow reversal. Detailed pipe-bend geometry and FEniCS settings are listed in Tables B.2 and B.3 of Section B.3; the corresponding U-bend setup is listed in Tables B.5 and B.6 of Section B.4. The Ansys grid-independence studies for the extracted turbulent designs are reported in Tables B.4 and B.7. Laminar-versus-turbulent comparisons are retained in Section C.2 and Section C.3 of Appendix C as supporting evidence for interpreting the optimized layouts.

## 6.2 Benchmark Strategy and Evaluation Metrics

Table 6.1 outlines the strategic role of the two benchmarks. The pipe bend receives the finite-difference sensitivity verification because it is the first final-configuration optimization case. The U-bend is then used to test whether the same methodology remains robust for a longer turning path with stronger flow reversal.

TABLE 6.1: Benchmark strategy used in Chapter 6. Detailed physical and numerical parameters are listed in Sections B.3 and B.4 of Appendix B.

| Case | Role in this thesis | Reference comparison | Validation scope |
|---|---|---|---|
| Pipe bend | Primary verification and optimization case at $V_f = 0.25$ | Pipe-bend setup and optimized fields from Bayat et al. [11] | Coordinate finite-difference check; FEniCS optimization; Ansys-SA and Ansys-SST validation; laminar-design comparison |
| U-bend | Companion curved-flow topology case at $V_f = 0.27$ | U-bend setup and optimized fields from Bayat et al. [11] | FEniCS optimization; Ansys-SA and Ansys-SST validation; laminar-design comparison |

### 6.2.1 Optimization and Validation Metrics

Throughout the optimization process, both benchmarks minimize the average inlet-pressure objective $J_p$ defined in Eq. 4.9. This pressure is averaged over the physical inlet boundary of the fixed non-design extension, where the inlet velocity is prescribed. The outlet pressure is fixed to zero, so reducing $J_p$ directly reduces the pressure level required to drive the prescribed inlet flow through the evolving topology. The optimization problem is still solved on a diffuse Brinkman domain, but once a design is extracted for Ansys Fluent, the Brinkman penalty is removed and the solid-fluid interface becomes a sharp wall.

Bayat et al. [11] report corresponding final average inlet-pressure objectives for their standard $k$-$\varepsilon$ computations in Table 7. Their "conventional" method is the density-based turbulent topology optimization formulation without the proposed implicit wall-functions: solid regions are imposed through Brinkman/body-force penalization, including damping of $k$ and $\varepsilon$ toward zero in solid material, on the same diffuse optimization mesh. It is therefore distinct from the body-fitted re-simulations with explicit wall-functions that they use for validation. Since the present thesis uses a wall-resolved SA model instead of $k$-$\varepsilon$, the conventional values from Bayat et al. are used only as scale comparisons in the individual benchmark discussions below.

To bridge this gap, the primary validation metric is the physical viscous dissipation integrated over the full exported CFD domain, including the non-design inlet, outlet, and fixed regions that are retained in the final geometry,

$$\Phi_D = \int_\Omega \frac{1}{2}\mu_{\text{eff}} \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right) : \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right) \, d\Omega, \tag{6.1}$$

where $\Omega$ denotes the complete sharp-geometry flow domain exported to Ansys. The effective viscosity is $\mu_{\text{eff}} = \mu + \mu_t$ for turbulent evaluations and $\mu_{\text{eff}} = \mu$ for laminar reference solves. Furthermore, because pressure drop is the most direct and practical engineering interpretation of these flow losses, $\Delta p$ is also reported from the Ansys validation runs. All two-dimensional dissipation quantities are evaluated per unit depth.

### 6.2.2 Wall-Distance and SA Penalization Conventions

The two benchmark computations use one wall-distance and penalty convention throughout: the reciprocal wall-distance equation is penalized by the power term in Eq. 4.5, and the SA working variable is suppressed by the power sink in Eq. 4.4. The wall-distance field is updated in the forward loop, but it is treated as a frozen coefficient during the adjoint sensitivity solve. Thus, the sensitivity verification below checks the implemented frozen derivative for the same practical assumption used by the optimizer.

## 6.3 Pipe-Bend Benchmark

The pipe-bend benchmark introduces a direct comparison with the turbulent topology optimization results of Bayat et al. [11]. While the reference study employs a standard $k$-$\varepsilon$ RANS model with wall functions, the present framework tackles the same problem using a wall-resolved SA model under the frozen-turbulence assumption. This comparison tests whether the fundamental topological solution, a streamlined and separation-avoiding bend, remains consistent across different turbulence closures.

### 6.3.1 Geometry and Boundary Conditions

The boundary conditions and geometric constraints replicate the setup shown in Fig. 10 of Bayat et al. [11], specifically targeting the $V_f = 0.25$ volume-fraction case. The inlet is placed on the left side of the non-design extension, the outlet is placed along the lower boundary, and the outlet pressure is fixed to zero. The optimization minimizes the average pressure on this prescribed inlet boundary required to drive the flow through the evolving bend.

### 6.3.2 Sensitivity Verification

Before interpreting the optimized topology, the frozen adjoint is checked directly on the pipe-bend configuration. This is the relevant local test for the final thesis setup because it uses the same topology-dependent SA and wall-distance penalties, inlet-pressure objective, and frozen-turbulence adjoint used during the optimization. No full turbulence-adjoint or wall-distance-adjoint comparison is made; the purpose is to compare the derivative supplied to MMA with objective changes obtained by re-solving the perturbed velocity-pressure state.

The check is performed at the initial pipe-bend design state, with base objective $J_0 = 1.8363345892928082$. In every perturbation, the SA working variable $\tilde{\nu}^0$ and reciprocal wall-distance field $G^0$ are held fixed. This matches the frozen-turbulence

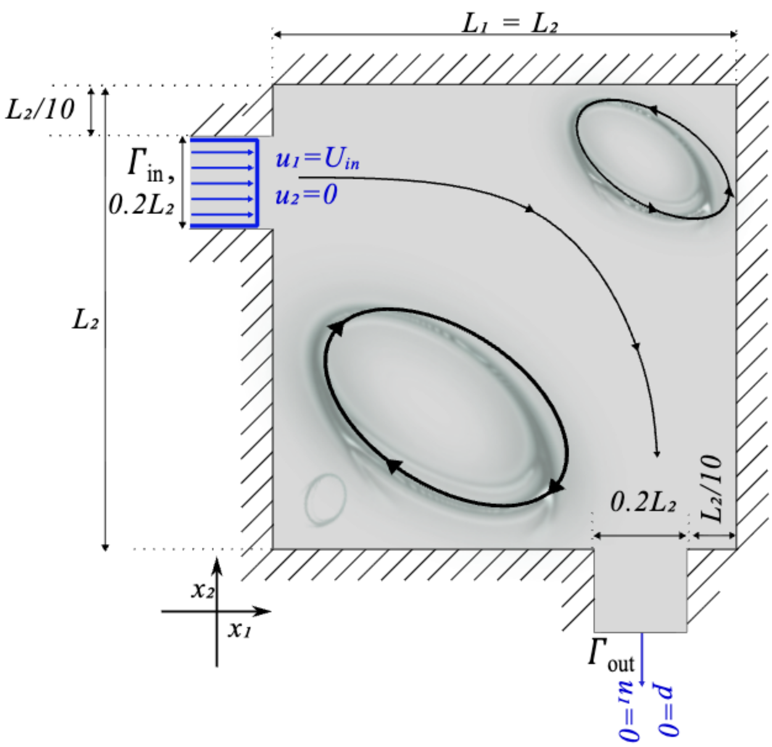



FIGURE 6.1: Geometry and boundary conditions for the pipe-bend benchmark from Bayat et al. [11]. The present evaluation targets the $V_f = 0.25$ case using the SA model.

assumption used by the adjoint: the flow state is re-solved, but the turbulence and wall-distance updates are not differentiated.

Five active $DG_0$ design cells are sampled deterministically from seed 13. For sampled cell $i$, the raw density vector $\boldsymbol{\gamma}$ is perturbed by $\pm\Delta h\,\mathbf{e}_i$, where $\mathbf{e}_i$ is the unit vector for that active design variable and $\Delta h = 10^{-6}$. The central finite-difference derivative is

$$S_i^{\mathrm{FD}} = \frac{J_p\big(\boldsymbol{\gamma} + \Delta h\,\mathbf{e}_i; \mathbf{u}^+, p^+, \tilde{\nu}^0, G^0\big) - J_p\big(\boldsymbol{\gamma} - \Delta h\,\mathbf{e}_i; \mathbf{u}^-, p^-, \tilde{\nu}^0, G^0\big)}{2\Delta h}. \tag{6.2}$$

The frozen-adjoint value for the same cell is denoted by $S_i^{\mathrm{fr}}$. The absolute and relative discrepancies are computed as

$$\Delta_i^{\mathrm{abs}} = \left|S_i^{\mathrm{fr}} - S_i^{\mathrm{FD}}\right|, \qquad \varepsilon_i^{\mathrm{fr}} = \frac{\Delta_i^{\mathrm{abs}}}{\max\big(\left|S_i^{\mathrm{fr}}\right|, \left|S_i^{\mathrm{FD}}\right|, 10^{-30}\big)}. \tag{6.3}$$

The coordinate check is mixed. The third sampled cell agrees to $3.84 \times 10^{-5}$ relative error and the fourth cell is within approximately 3.05%, but the remaining sampled cells show relative discrepancies between 24.8% and 67.9%. Thus, the coordinate data should be read as a Dilgen-style diagnostic of the current frozen-gradient implementation rather than as a uniformly successful finite-difference verification.

Table 6.2: Coordinate sensitivity check for the frozen pipe-bend configuration. Values are taken from `SensitivityVerificationTable.tsv`.

| **Cell** | **Finite difference** $S_i^{\mathrm{FD}}$ | **Frozen adjoint** $S_i^{\mathrm{fr}}$ | **Absolute error** $\Delta_i^{\mathrm{abs}}$ | **Relative error** $\varepsilon_i^{\mathrm{fr}}$ |
|---|---|---|---|---|
| $CV_1$ | $-2.7041 \times 10^{-5}$ | $-5.2364 \times 10^{-5}$ | $2.5323 \times 10^{-5}$ | $4.8359 \times 10^{-1}$ |
| $CV_2$ | $-2.1494 \times 10^{-6}$ | $-2.8564 \times 10^{-6}$ | $7.0704 \times 10^{-7}$ | $2.4752 \times 10^{-1}$ |
| $CV_3$ | $-1.6641 \times 10^{-4}$ | $-1.6641 \times 10^{-4}$ | $6.3979 \times 10^{-9}$ | $3.8445 \times 10^{-5}$ |
| $CV_4$ | $-1.7281 \times 10^{-5}$ | $-1.7824 \times 10^{-5}$ | $5.4339 \times 10^{-7}$ | $3.0487 \times 10^{-2}$ |
| $CV_5$ | $2.3808 \times 10^{-5}$ | $7.4164 \times 10^{-5}$ | $5.0356 \times 10^{-5}$ | $6.7898 \times 10^{-1}$ |

The check is intentionally limited to one representative case because it exercises the same frozen derivative pathway used by the U-bend. Repeating the same diagnostic on the U-bend would test the same algebraic implementation path rather than a fundamentally different adjoint formulation. Since there is no independent reference field for this SA frozen derivative, a spatial $dJ_p/d\gamma$ contour is not used as a verification metric; the relevant evidence is the sampled finite-difference comparison reported above.

### 6.3.3 Optimization Results

Figure 6.2 illustrates the projected density field $\bar{\gamma}$ and the corresponding velocity magnitude. Despite the differing turbulence models, it is anticipated that the FEniCS-SA optimizer will converge on a broadly similar routing strategy as the $k$-$\varepsilon$ reference. The primary high-speed flow path should bypass the sharpest corners to minimize momentum loss, although subtle differences in the treatment of the boundary layer may result in variations in the exact gray-scale density distribution near the walls.

**Placeholder**

Projected density field $\bar{\gamma}$, $V_f = 0.25$

(a) Projected density. Final objective $J_p = 0.3023$.

**Placeholder**

Velocity-magnitude contours

(b) Velocity magnitude.

Figure 6.2: FEniCS-SA optimization result for the pipe-bend benchmark at $V_f = 0.25$.

For the same $V_f = 0.25$, $Re = 5{,}000$ operating point, Bayat et al. [11] report

$J_p = 0.4023$ for their conventional $k$-$\varepsilon$ topology optimization. The present FEniCS-SA value is therefore of the same order and lower for this run, but the difference should be interpreted as indicative rather than a direct model-to-model validation because the turbulence closures differ.

**Placeholder**

Objective function vs.
optimization iteration

(A) Objective function.

**Placeholder**

Volume constraint vs.
optimization iteration

(B) Volume constraint.

FIGURE 6.3: Optimization convergence histories for the pipe-bend benchmark.

### 6.3.4 CFD Validation

The structural robustness of the extracted pipe-bend topology is verified in Table 6.3. Because the SST $k$-$\omega$ model is notably more sensitive to adverse pressure gradients than SA, evaluating the extracted geometry under both closures serves as a rigorous stress test for the optimized shape.

TABLE 6.3: CFD validation metrics for the extracted pipe-bend topology.

| **Solver** | **Model** | $\Phi_D$ **[W/m]** | $\Delta p$ **[Pa]** | **Comment** |
|---|---|---|---|---|
| FEniCS diffuse design | SA | [TBD] | [TBD] | Optimization state |
| Ansys sharp geometry | SA | [TBD] | [TBD] | Cross-code comparison |
| Ansys sharp geometry | SST $k$-$\omega$ | [TBD] | [TBD] | Turbulence-model sensitivity |

### 6.3.5 Comparison with Laminar Design

The laminar-design comparison is reported in Section C.2 of Appendix C. The expected distinction is that a laminar bend optimized at $Re = 1$ can minimize wall length and viscous shear in the creeping-flow limit, but may produce a sharper or less separation-aware turn when evaluated at the target turbulent Reynolds number. The comparison is therefore interpreted under identical turbulent operating conditions, rather than by comparing the laminar and turbulent optimization objectives directly.

## 6.4 U-Bend Benchmark

The U-bend is the companion benchmark to the pipe bend. It uses the same frozen-SA topology optimization framework, but the geometry forces a 180° return path between two ports on the same side of the domain. This makes the case useful for checking whether the optimizer can form a smooth turbulent return duct without relying on the shorter 90° routing available in the pipe-bend case.

### 6.4.1 Geometry and Boundary Conditions

The geometry and boundary conditions follow Fig. 11 and Table 4 of Bayat et al. [11]. The square design region has side length $L$, with a non-design lead duct of length $0.2L$ on the left. The inlet and outlet ports both have height $0.2L$; the top port acts as the inlet and the lower port acts as the pressure outlet. A fixed internal separator bar of thickness $0.10L$ extends from the lead duct into the design domain, forcing the flow to turn around its rounded tip. The target volume fraction is $V_f = 0.27$, and the Reynolds number is $Re = 5{,}000$ based on the port height and maximum inlet velocity.

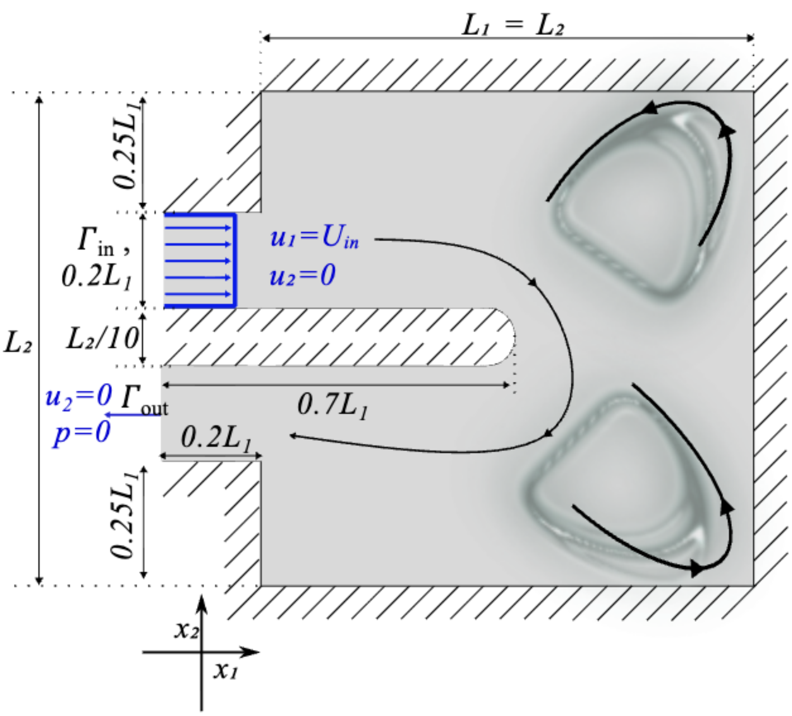



Figure 6.4: Geometry and boundary conditions for the U-bend benchmark from Bayat et al. [11].

### 6.4.2 Optimization Results

Figure 6.5 reports the final projected density and velocity-magnitude field for the U-bend. The main feature to assess is whether the design forms a rounded return passage around the separator tip, rather than a short path with tight local curvature. Because the objective is average inlet pressure, the optimized layout should reduce the pressure required to maintain the imposed inlet flow while respecting the prescribed volume fraction.

**Placeholder**

Projected density field $\bar{\gamma}$, $V_f = 0.27$

(A) Projected density. Final objective $J_p = 1.1792$.

**Placeholder**

Velocity-magnitude contours

(B) Velocity magnitude.

FIGURE 6.5: FEniCS-SA optimization result for the U-bend benchmark at $V_f = 0.27$.

For the matching $V_f = 0.27$, $Re = 5{,}000$ U-bend case, the conventional $k$-$\varepsilon$ result reported by Bayat et al. [11] is $J_p = 0.8327$. The present FEniCS-SA run gives a higher final inlet-pressure objective, so the comparison suggests a less favorable pressure-loss level for this topology while still remaining a qualitative comparison across different turbulence models.

**Placeholder**

Objective function vs. optimization iteration

(A) Objective function.

**Placeholder**

Volume constraint vs. optimization iteration

(B) Volume constraint.

FIGURE 6.6: Optimization convergence histories for the U-bend benchmark.

### 6.4.3 CFD Validation

The extracted U-bend topology is validated with the same sharp-geometry workflow as the pipe bend. The Ansys-SA solve checks cross-code consistency for the same turbulence closure family, while the Ansys SST $k$-$\omega$ solve tests whether the design remains robust under a two-equation model that is more sensitive to separation and adverse pressure gradients.

Table 6.4: CFD validation metrics for the extracted U-bend topology.

| Solver | Model | $\Phi_D$ [W/m] | $\Delta p$ [Pa] | Comment |
|---|---|---|---|---|
| FEniCS diffuse design | SA | [TBD] | [TBD] | Optimization state |
| Ansys sharp geometry | SA | [TBD] | [TBD] | Cross-code comparison |
| Ansys sharp geometry | SST $k$-$\omega$ | [TBD] | [TBD] | Turbulence-model sensitivity |

### 6.4.4 Comparison with Laminar Design

The laminar-design comparison is reported in Section C.3 of Appendix C. The laminar reference uses the same U-bend mesh, inlet and outlet ports, separator geometry, and volume fraction, but it is optimized at $Re = 1$ without the SA turbulence model. The extracted laminar and turbulent layouts are then evaluated under the same target turbulent operating conditions. This comparison is expected to show whether the laminar optimizer creates an overly tight return path around the separator tip, while the turbulent optimizer allocates fluid volume to reduce separation and momentum loss through the 180° turn.

## 6.5 Conclusion

This chapter centers the final thesis evidence on the pipe-bend and U-bend benchmarks from Bayat et al. [11]. The sensitivity verification is performed on the same penalized wall-distance, topology-penalized SA, frozen-turbulence formulation used for the optimization, so the coordinate finite-difference comparison directly checks the derivative supplied to MMA. The pipe-bend and U-bend results then test the same framework on two distinct curved-flow layouts.

The independent CFD validation using Ansys Fluent is therefore interpreted as a sharp-geometry check of the optimized designs, not merely as another internal FEniCS result. Testing the extracted topologies with both SA and SST $k$-$\omega$ quantifies how much of the optimized performance survives a change in turbulence closure. Together with the laminar comparisons, this chapter supports the thesis argument that high-Reynolds turbulence physics must be included directly in the optimization loop when designing viable internal-flow topologies.

# Chapter 7

# Conclusion and Future Work

## 7.1 Conclusion

The primary objective of this thesis was to develop a stable, continuous turbulent topology optimization framework within FEniCS, verify its forward-flow and sensitivity behavior, and critically evaluate the physical credibility of its resulting designs. The final thesis structure deliberately separates methodology from evidence: Chapters 3 and 4 define the solver and optimization methods, Chapter 5 verifies the standalone SA flow solver, and Chapter 6 evaluates the pipe-bend and U-bend topology optimization benchmarks from Bayat et al. [11] with independent CFD re-analysis.

Addressing the **first research question**, the theoretical limitations and computational advantages of the Spalart-Allmaras (SA) turbulence model were assessed. While its one-equation, wall-resolved formulation makes it highly tractable for continuous adjoint frameworks, its reliance on the linear Boussinesq approximation fundamentally limits its accuracy in resolving anisotropic Reynolds stresses—particularly in regimes dominated by high streamline curvature and separation.

Regarding the **second research question**, the numerical integration of this highly non-linear SA model into the FEniCS finite-element environment was successfully achieved. By deploying a decoupled Picard iteration scheme, an Incremental Pressure Correction Scheme (IPCS) for the Navier-Stokes equations, and a relaxed Eikonal equation for wall-distance computation, the forward fluid state was robustly driven to equilibrium. The channel and U-bend verification cases in Chapter 5 show that this implementation reproduces the relevant Ansys-SA flow behavior with sufficient accuracy for the subsequent optimization campaign.

Building on this foundation, the mathematical framework for topology optimization was constructed. The implementation confirmed that the "frozen turbulence" adjoint assumption, which treats the turbulent eddy viscosity and reciprocal wall-distance fields as fixed coefficients during gradient assembly, provides a stable and computationally efficient optimization engine. This assumption is approximate, and the pipe-bend finite-difference check in Section 6.3.2 of Chapter 6 provides a Dilgen-style coordinate diagnostic of the implemented frozen derivative for the configuration used in the final optimization campaign.

Finally, answering the **third research question**, the framework was subjected to strict cross-code CFD validation. The pipe bend served as the primary verification and optimization case, while the U-bend provided a companion curved-flow topology benchmark with stronger flow reversal. In both cases, the diffuse FEniCS-SA topology is extracted as a sharp geometry and re-analyzed in Ansys Fluent. The laminar-design comparisons further support the engineering motivation for turbulent optimization by showing why laminar-optimized layouts can perform poorly at high Reynolds numbers.

Re-evaluating the extracted geometries using the Ansys SST $k-\omega$ model quantifies the sensitivity of the optimized designs to the underlying turbulence physics. This confirms that topologies optimized under isotropic, one-equation assumptions must be checked carefully when subjected to secondary flows, curvature effects, and anisotropic shear layers captured more strongly by two-equation models.

## 7.2 Limitations

An objective evaluation of this methodology reveals several important limitations that must be acknowledged:

1. **The Frozen Adjoint Approximation:** By ignoring the derivative of the eddy viscosity with respect to the design field, the optimizer cannot mathematically anticipate future turbulence production. In flows dominated by massive separation, this can blind the optimizer to the downstream hydrodynamic consequences of upstream geometric changes, potentially leading to sub-optimal local minima.

2. **Geometry Extraction Uncertainties:** The transition from a continuous, Brinkman-penalized density field ($\gamma \in [0, 1]$) to a Boolean CAD geometry introduces inherent ambiguity. The specific iso-value threshold chosen to extract the fluid volume dictates the final wall placement. Consequently, the performance discrepancies observed between FEniCS and Ansys are inextricably linked to this manual geometry extraction process, making it difficult to isolate pure solver physics from thresholding effects.

3. **Two-Dimensional Simplification:** The benchmark geometries evaluated in this thesis were restricted to 2D planar domains. Because turbulent phenomena—particularly curvature-induced secondary flows like Dean vortices and turbulent energy cascades—are inherently three-dimensional, the 2D assumption artificially suppresses complex mixing mechanisms that would inevitably occur in physical prototypes.

4. **Lack of Experimental Validation:** While independent cross-code verification against commercial finite-volume solvers strongly corroborates the numerical implementation, it is not a substitute for physical validation. The true operational performance of these algorithmically generated turbulent manifolds

requires future validation against experimental pressure-drop and flow-field measurements.

## 7.3 Future Work

The verified turbulent fluid mechanics framework developed in this thesis establishes a mathematically stable foundation for several advanced avenues of future research:

### 7.3.1 Conjugate Heat Transfer (CHT) and Thermal Management

The ultimate engineering application for high-Reynolds topology optimization is not merely fluid routing, but active thermal management. Historically, the immense computational cost of resolving Conjugate Heat Transfer (CHT) has forced optimization frameworks to rely on laminar approximations (e.g., natural convection as demonstrated by Alexandersen et al. [2]) or simplified flow proxies such as Darcy models [49]. Even when high-resolution, density-based solvers are successfully deployed for microelectronics cooling, they typically circumvent the complexities of turbulent momentum transport [10].

However, the necessity of capturing true turbulent convection in high-performance heat sinks has driven recent advancements in the literature. Emerging efforts have introduced turbulence into thermal topology optimization using extended finite element (XFEM) level-set approaches with the SA model [29], computationally reduced multi-layer thermofluid approximations [48], and standard density-based frameworks coupled to two-equation $k - \epsilon$ closures [38].

With the turbulent momentum and SA transport equations now stably implemented and verified within the current FEniCS architecture, this framework is perfectly positioned to contribute to this cutting-edge research front. Extending the current methodology to CHT will require implementing a turbulent Prandtl number ($Pr_t$) approach to model the turbulent heat flux vector, alongside deriving the corresponding thermal adjoint sensitivities to minimize peak temperatures or maximize overall heat transfer coefficients. Furthermore, the capacity for optimizing two-fluid heat exchangers—currently dominated by laminar assumptions [24]—could be significantly enhanced by adapting this turbulent solver.

### 7.3.2 Three-Dimensional Topologies

As noted in the limitations, turbulent secondary flows are inherently 3D phenomena. To fully evaluate the geometric limits of the turbulence models and design physically realizable manifolds, the current 2D FEniCS implementation must be extended to three dimensions. While the governing equations and adjoint derivations remain mathematically identical, the computational cost of resolving the boundary layers ($y^+ \leq 1$) across a 3D topology optimization grid will require significant advancements in parallel processing and highly efficient linear solvers (e.g., Algebraic Multigrid preconditioners).

### 7.3.3 Adjoint Model Hierarchies

This thesis deliberately uses the frozen-turbulence adjoint as the robust baseline. Future frameworks should investigate stabilized ways of reintroducing selected turbulence-production sensitivities, especially in separated flows where the Constant Eddy Viscosity assumption is least accurate. Higher-fidelity two-equation models, such as $k - \omega$, should also be investigated directly within the continuous adjoint topology optimization loop to mitigate the curvature limitations identified in Chapter 6.

# Appendices

# Appendix A

# Configuration and Grid Independence of the Spalart-Allmaras Verification Cases

To ensure complete scientific reproducibility and to prevent the disruption of the narrative flow in Chapter 3, all comprehensive configuration matrices, domain truncation proofs, and spatial grid independence studies for the fluid benchmarks are documented in this appendix.

## A.1 Verification Benchmark 1: Channel Flow

### A.1.1 Physical and Geometric Parameters

The first verification benchmark consists of a pressure-driven periodic 2D channel. To align with fundamental turbulence research conventions, the characteristic length is defined as the full channel height ($H$), and the Reynolds number ($Re_H$) is computed strictly based on this geometric parameter rather than the hydraulic diameter. The complete physical setup is detailed in Table A.1.

### A.1.2 Numerical Simulation Parameters

The channel flow was resolved using an Incremental Pressure Correction Scheme (IPCS) embedded within a Picard fixed-point iteration loop. The exact numerical solver configurations, convergence tolerances, and relaxation factors utilized in FEniCS are summarized in Table A.2.

### A.1.3 Spatial Verification and Grid Independence

To guarantee that the numerical results were independent of the spatial discretization, a formal grid independence study was conducted. Three distinct boundary-

TABLE A.1: Geometric, physical, and boundary-condition parameters for the periodic channel flow benchmark.

| Parameter | Channel flow setting |
|---|---|
| Benchmark type | Pressure-driven, streamwise-periodic 2D channel |
| Domain dimensions | Length $L = 2.0$ m; full channel height $H = 2.0$ m |
| Characteristic length | Full channel height $H = 2.0$ m |
| Reference velocity | $U_{\text{ref}} = 18.5$ m/s, utilized to define the target Reynolds number |
| Kinematic viscosity | $\nu = 1.81818 \times 10^{-3}$ m$^2$/s |
| Reynolds number | $Re_H \approx 2.03 \times 10^4$ |
| Body force | $\mathbf{f} = (0, 0)$ m/s$^2$ |
| Driving condition | Fixed pressure drop from $p = 2$ Pa to $p = 0$ Pa across the periodic streamwise direction |
| Velocity boundary conditions | Periodic velocity between inlet/outlet planes; no-slip on top and bottom walls |
| Pressure boundary conditions | $p = 2$ Pa at the upstream pressure plane; $p = 0$ Pa at the downstream pressure plane |
| SA boundary conditions | Periodic $\tilde{\nu}$ between inlet/outlet planes; $\tilde{\nu} = 0$ at walls |
| Initial fields | $\mathbf{u}_0 = (18.5, 0)$ m/s, $p_0 = 2$ Pa, $\tilde{\nu}_0 = 5.0 \times 10^{-3}$ m$^2$/s |
| Verification mesh | `Fine_FEniCS`; first layer $\Delta y_1 = 1.709 \times 10^{-3}$ m |

layer inflated meshes (Coarse, Medium, and Fine) were evaluated. The geometric properties, node counts, and resolution parameters of this mesh hierarchy are explicitly detailed in Table A.3. Crucially, the first-layer height was kept constant across all grids to strictly enforce the $y^+ \approx 1$ requirement, while the bulk and inflation-layer resolutions were systematically refined.

Visual confirmation of this spatial convergence is provided by the wall-normal velocity profiles plotted in Figure A.1. The `Fine_FEniCS` grid was consequently selected for the final verification against the Ansys Fluent baseline, while the coarser grids demonstrate that the wall-normal velocity profile is already asymptotically converged.

Furthermore, to ensure a rigorous cross-code validation, an equivalent grid independence study was performed for the commercial Ansys Fluent baseline. The Ansys domain was discretized using a comparable mapped mesh hierarchy, maintaining a strict $y^+ \approx 1$ resolution at the walls. The properties of these meshes are documented in Table A.4. The wall-normal velocity profiles extracted from the Ansys solver, illustrated in Figure A.2, exhibited identical asymptotic convergence, confirming that the baseline reference data is mathematically robust and completely independent of spatial discretization errors.

Table A.2: Numerical simulation parameters for the FEniCS steady Spalart-Allmaras channel benchmark.

| Parameter | Symbol | Setting |
|---|---|---|
| Velocity space | $\mathbf{u}$ | Vector CG2, periodic in streamwise direction |
| Pressure space | $p$ | Scalar CG1 |
| SA working-variable space | $\tilde{\nu}$ | Scalar CG1, periodic in streamwise direction |
| Quadrature degree | – | 2 |
| Maximum outer Picard steps | $N_{\text{Picard,max}}$ | 200 |
| SA solves per Picard step | – | 1 |
| Outer velocity tolerance | $\epsilon_{\mathbf{u}}$ | $1.0 \times 10^{-6}$ |
| Outer pressure tolerance | $\epsilon_p$ | $1.0 \times 10^{-6}$ |
| Outer SA tolerance | $\epsilon_{\tilde{\nu}}$ | $1.0 \times 10^{-6}$ |
| IPCS pseudo-time step | $\Delta t$ | $5.0 \times 10^{-3}$ s |
| Maximum IPCS steps per Picard step | – | 500 |
| IPCS velocity tolerance | – | $1.0 \times 10^{-6}$ |
| IPCS pressure tolerance | – | $1.0 \times 10^{-6}$ |
| Velocity relaxation factor | $\alpha_u$ | 0.3 |
| Pressure relaxation factor | $\alpha_p$ | 0.3 |
| SA relaxation factor | $\alpha_{\tilde{\nu}}$ | 0.3 |
| SA lower bound | $\tilde{\nu}_{\min}$ | $1.0 \times 10^{-12}$ |
| Wall-distance equation | – | Relaxed Eikonal, $\sigma_w = 0.1$, $G_0 = 20$ |
| Wall-distance Newton settings | – | Relative tolerance $1.0 \times 10^{-8}$; absolute tolerance $1.0 \times 10^{-10}$; maximum 80 iterations; relaxation 0.5 |
| Linear solver strategy | – | IPCS blocks: direct MUMPS; SA transport: default solver |

Table A.3: Wall-resolved channel mesh hierarchy used for the FEniCS grid-independence study. The first-layer height is kept fixed to maintain $y^+ \approx 1$, while the streamwise resolution, wall-normal resolution, and inflation-layer resolution are refined.

| Parameter | Coarse_FEniCS | Medium_FEniCS | Fine_FEniCS |
|---|---|---|---|
| Streamwise cells | 10 | 20 | 40 |
| Wall-normal cells | 42 | 51 | 68 |
| First-layer height $\Delta y_1$ [m] | $1.709 \times 10^{-3}$ | $1.709 \times 10^{-3}$ | $1.709 \times 10^{-3}$ |
| Minimum wall-normal spacing [m] | $1.709 \times 10^{-3}$ | $1.709 \times 10^{-3}$ | $1.709 \times 10^{-3}$ |
| Maximum wall-normal spacing [m] | $7.707 \times 10^{-2}$ | $8.454 \times 10^{-2}$ | $7.589 \times 10^{-2}$ |
| Inflation layers per wall | 10 | 14 | 18 |
| Growth ratio | 1.45 | 1.35 | 1.25 |
| Vertices | 473 | 1,092 | 2,829 |
| Triangular elements | 840 | 2,040 | 5,440 |

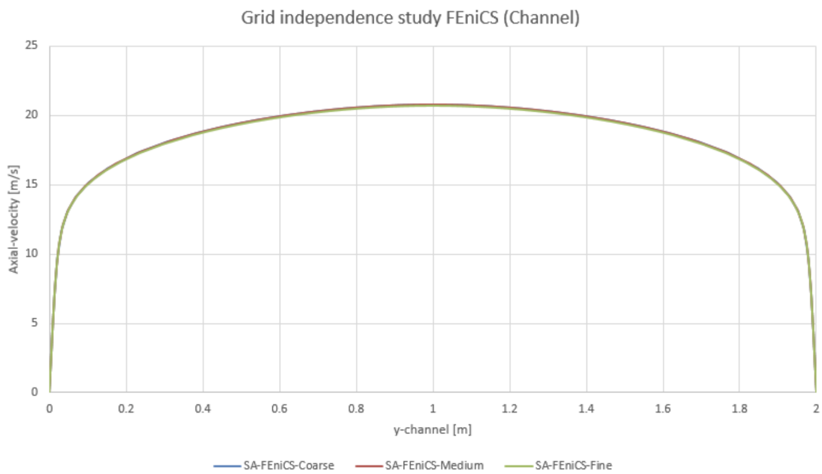



FIGURE A.1: Wall-normal velocity profiles for the Coarse, Medium, and Fine grids in the periodic channel flow benchmark. The overlapping profiles demonstrate asymptotic spatial convergence.

TABLE A.4: Wall-resolved channel mesh hierarchy used for the Ansys grid-independence study.

| **Parameter** | **Coarse_Ansys** | **Medium_Ansys** | **Fine_Ansys** |
|---|---|---|---|
| Streamwise cells | – | – | – |
| Wall-normal cells | – | – | – |
| First-layer height $\Delta y_1$ [m] | – | – | – |
| Minimum wall-normal spacing [m] | – | – | – |
| Maximum wall-normal spacing [m] | – | – | – |
| Inflation layers per wall | – | – | – |
| Growth ratio | – | – | – |
| Vertices | – | – | – |
| Elements | – | – | – |

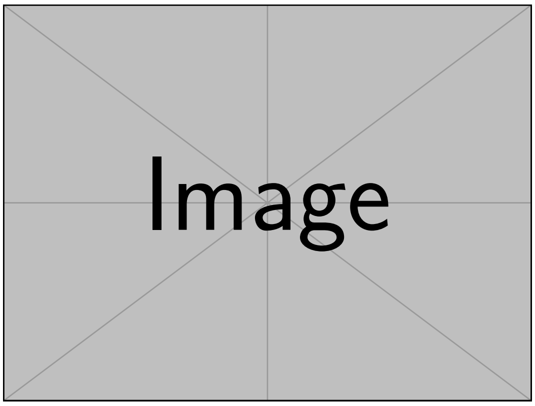



Figure A.2: Wall-normal velocity profiles for the Coarse, Medium, and Fine grids in the Ansys periodic channel flow benchmark, demonstrating spatial convergence of the baseline data.

## A.2 Verification Benchmark 2: U-Bend Geometry

### A.2.1 Physical and Geometric Parameters

The second verification benchmark evaluates a highly convective U-bend. Because this FEniCS domain is a 2D planar approximation of the symmetry mid-plane of the 3D Ansys VMFL048 case, the 3D physical pipe diameter ($D$) is explicitly retained as the characteristic length scale. This ensures the Reynolds ($Re_D$) and Dean ($De$) numbers remain non-dimensionally consistent with the commercial baseline. The configuration is detailed in Table A.5.

TABLE A.5: Geometric, physical, and boundary-condition parameters for the U-bend benchmark.

| **Parameter** | **U-bend setting** |
|---|---|
| Benchmark type | Truncated 2D symmetry-plane U-bend based on Ansys VMFL048 |
| Domain dimensions | Pipe diameter $D = 0.028$ m; centreline bend radius $R_c = 0.125$ m; straight-leg length $30D = 0.840$ m |
| Characteristic length | Physical pipe diameter $D = 0.028$ m |
| Reference / inlet velocity | Uniform prescribed inlet velocity $\mathbf{u}_{in} = (0, -1.42)$ m/s |
| Kinematic viscosity | $\nu = 8.9 \times 10^{-7}$ m$^2$/s |
| Reynolds number | $Re_D \approx 4.47 \times 10^4$ |
| Dean number | $De \approx 1.50 \times 10^4$ |
| Body force | $\mathbf{f} = (0, 0)$ m/s$^2$ |
| Driving condition | Prescribed inlet velocity and zero relative static pressure at the outlet |
| Velocity boundary conditions | No-slip on all curved and straight pipe walls; natural velocity condition at pressure outlet |
| Pressure boundary conditions | $p = 0$ Pa at outlet; pressure left free at inlet and walls |
| SA boundary conditions | $\tilde{\nu}_{in} = 9.34\times10^{-5}$ m$^2$/s at inlet; $\tilde{\nu} = 0$ at walls; natural condition at outlet |
| Initial fields | $\mathbf{u}_0 = (0, 0)$ m/s, $p_0 = 0$ Pa, $\tilde{\nu}_0 = \tilde{\nu}_{in}$ |
| Verification mesh | `Medium_FEniCS`; first layer $\Delta y_1 = 1.20 \times 10^{-5}$ m |

### A.2.2 Numerical Simulation Parameters

Due to the adverse pressure gradients and secondary flow separation induced by the streamline curvature, the U-bend represents a significantly more demanding numerical challenge than the channel. Consequently, the FEniCS solver required a smaller pseudo-time step, tighter under-relaxation, and advanced iterative preconditioners to maintain stability, as detailed in Table A.6.

### A.2.3 Domain Truncation Study

The original 3D Ansys VMFL048 benchmark utilizes highly extended straight legs measuring 1.555 m to ensure fully developed flow. To reduce computational over-

Table A.6: Numerical simulation parameters for the highly convective FEniCS U-bend benchmark.

| **Parameter** | **Symbol** | **Setting** |
|---|---|---|
| Velocity space | $\mathbf{u}$ | Vector CG2 |
| Pressure space | $p$ | Scalar CG1 |
| SA working-variable space | $\tilde{\nu}$ | Scalar CG1 |
| Quadrature degree | – | 4 |
| Maximum outer Picard steps | $N_{\text{Picard,max}}$ | 180 |
| SA solves per Picard step | – | 1 |
| Outer velocity tolerance | $\epsilon_{\mathbf{u}}$ | $1.0 \times 10^{-4}$ |
| Outer pressure tolerance | $\epsilon_p$ | $1.0 \times 10^{-4}$ |
| Outer SA tolerance | $\epsilon_{\tilde{\nu}}$ | $1.0 \times 10^{-6}$ |
| IPCS pseudo-time step | $\Delta t$ | $1.0 \times 10^{-4}$ s |
| Maximum IPCS steps per Picard step | – | 180 |
| IPCS velocity tolerance | – | $1.0 \times 10^{-4}$ |
| IPCS pressure tolerance | – | $2.0 \times 10^{-3}$ |
| Velocity relaxation factor | $\alpha_u$ | 0.3 |
| Pressure relaxation factor | $\alpha_p$ | 0.1 |
| SA relaxation factor | $\alpha_{\tilde{\nu}}$ | 0.01 |
| SA lower bound | $\tilde{\nu}_{\min}$ | $1.0 \times 10^{-12}$ |
| Wall-distance equation | – | Relaxed Eikonal, $\sigma_w = 0.1$, $G_0 = 20$ |
| Wall-distance Newton settings | – | Relative tolerance $1.0 \times 10^{-8}$; absolute tolerance $1.0 \times 10^{-10}$; maximum 80 iterations; relaxation 0.5 |
| Linear solver strategy | – | IPCS blocks: GMRES with ILU velocity/correction preconditioning and Hypre-AMG pressure preconditioning; SA transport: default solver |

head for subsequent optimization stages, these legs were truncated to $30D$ (0.840 m). A preliminary sensitivity study was conducted in Ansys Fluent, comparing the full-length and truncated geometries. To mathematically prove that this reduction does not artificially alter the flow physics, the cross-sectional velocity profiles at the critical bend exit were extracted for both domains, as plotted in Figure A.3. The complete overlap of the profiles confirms that the $30D$ truncation successfully isolated the curvature physics without artificially impacting the boundary layer separation.

### A.2.4 Spatial Verification and Grid Independence

Managing the highly stretched, non-orthogonal inflation layers around the curved arcs of the U-bend required custom mesh generation via Gmsh. A spatial grid independence study was conducted across three refined unstructured grids. The node counts, element densities, and inflation layer parameters for these generated meshes are explicitly documented in Table A.7. Similar to the channel benchmark, the wall-

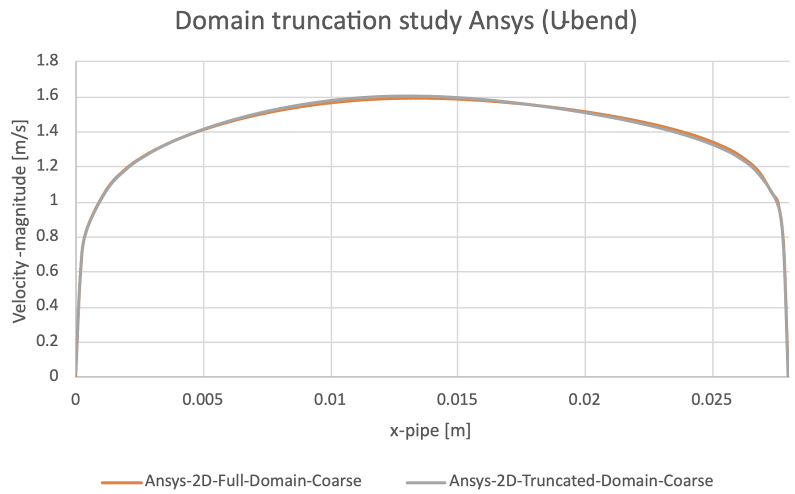



FIGURE A.3: Velocity profile comparison at the U-bend exit between the full Ansys domain (1.555 m legs) and the truncated computational domain ($30D$ legs). The complete overlap confirms the validity of the truncated geometry.

adjacent element height was strictly maintained at $1.20 \times 10^{-5}$ m to satisfy the low-Reynolds SA wall constraints, while the internal element sizing was systematically refined.

TABLE A.7: Wall-resolved U-bend mesh hierarchy used for the FEniCS grid-independence study. The first-layer height is kept fixed to maintain $y^+ \approx 1$, while the bulk element size and inflation-layer resolution are refined.

| **Parameter** | **Coarse_FEniCS** | **Medium_FEniCS** | **Fine_FEniCS** |
|---|---|---|---|
| Bulk element size [m] | $2.80 \times 10^{-3}$ | $1.40 \times 10^{-3}$ | $8.00 \times 10^{-4}$ |
| First-layer height $\Delta y_1$ [m] | $1.20 \times 10^{-5}$ | $1.20 \times 10^{-5}$ | $1.20 \times 10^{-5}$ |
| Inflation layers | 10 | 14 | 18 |
| Growth ratio | 1.45 | 1.30 | 1.25 |
| Inflation thickness [m] | $1.07 \times 10^{-3}$ | $1.53 \times 10^{-3}$ | $2.62 \times 10^{-3}$ |
| Vertices | 23,751 | 73,978 | 182,450 |
| Triangular elements | 45,998 | 144,952 | 359,646 |

Convergence was visually verified by extracting the velocity profiles directly at the bend exit, a highly sensitive region for curvature-induced separation, as illustrated in Figure A.4. The chosen `Medium_FEniCS` grid perfectly captured the velocity gradients near the wall and provided sufficient isotropic resolution in the bulk flow to resolve the separation bubbles without numerical instability.

Similarly, to guarantee the integrity of the cross-code validation in this highly convective regime, an equivalent mesh independence study was executed for the Ansys Fluent baseline. Using a parallel unstructured meshing strategy with strict $y^+ \approx 1$ boundary layer inflation (documented in Table A.8), the Ansys velocity

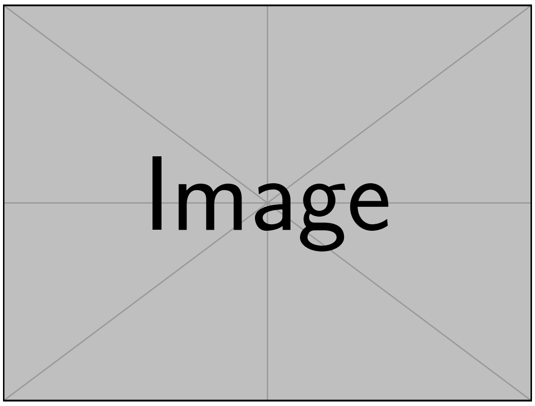



Figure A.4: Velocity profiles at the U-bend exit for the Coarse, Medium, and Fine grids in FEniCS. The profiles demonstrate spatial convergence, justifying the selection of the Medium grid for the final verification.

profiles at the bend exit, shown in Figure A.5, were verified to be spatially converged. This ensures that the commercial baseline reference data is independent of grid resolution.

Table A.8: Wall-resolved U-bend mesh hierarchy used for the Ansys grid-independence study.

| **Parameter** | **Coarse_Ansys** | **Medium_Ansys** | **Fine_Ansys** |
|---|---|---|---|
| Bulk element size [m] | – | – | – |
| First-layer height $\Delta y_1$ [m] | – | – | – |
| Inflation layers | – | – | – |
| Growth ratio | – | – | – |
| Inflation thickness [m] | – | – | – |
| Vertices | – | – | – |
| Elements | – | – | – |

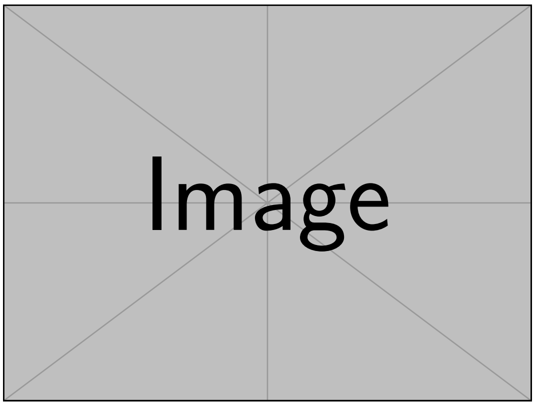



Figure A.5: Velocity profiles at the U-bend exit for the Coarse, Medium, and Fine Ansys grids, demonstrating spatial convergence of the commercial baseline.

# Appendix B

# Configuration and Grid Independence of the Alexandersen Benchmarks

This appendix collects the detailed setup information for the Alexandersen topology optimization benchmarks used in Chapter 6. Only the Alexandersen pipe bend and Alexandersen U-bend are included. For each case, the appendix records the geometry and physical parameters, the FEniCS simulation and optimization parameters, and the Ansys Fluent mesh-independence tables used for sharp-geometry CFD validation.

## B.1 Validation and Meshing Strategy

The FEniCS meshes used during topology optimization are treated as fixed optimization discretizations rather than as a separate mesh-convergence hierarchy. This distinction is important because refining the FEniCS mesh would also change the $DG_0$ design space, the filter stencil, the number of MMA design variables, and the diffuse Brinkman interface location. The pipe-bend optimization uses a rectangular-coordinate triangular mesh with near-wall clustering based on a smooth-wall $y^+ \approx 1$ estimate at the prescribed inlet Reynolds number. The U-bend optimization uses the corresponding wall-resolved triangular Gmsh mesh, with a near-wall size field based on the same $y^+ \approx 1$ estimate and a far-field target size of $h_{\max} = 0.007L$ used to set the density-filter scale. In both cases, the mesh is chosen to provide sufficient wall-resolved SA and design resolution for the optimization problem; quantitative grid-convergence evidence is instead reported for the extracted sharp geometries in Ansys Fluent.

For each extracted turbulent topology optimization design, the Ansys validation mesh is refined through coarse, medium, and fine resolutions. The reported validation metrics in Chapter 6 should be taken from a mesh for which both viscous dissipation and pressure drop have become insensitive to further refinement. The selected mesh should be reported explicitly for each case.

Both SA and SST $k$-$\omega$ validation runs use the same extracted geometry, inlet condition, outlet condition, and fluid properties. The mesh-independence comparison is reported for both turbulence models because near-wall behavior and separated-flow predictions may respond differently to grid refinement.

## B.2 Shared Alexandersen Topology Optimization Conventions

Both Alexandersen cases use the same turbulence-topology coupling conventions. Topology-created solids act as walls for the SA working-variable penalty and the reciprocal wall-distance solve. The design density source is therefore `design`, the wall-distance mode is `reciprocal_penalized`, and both the SA and wall-distance penalties use the power-law solid indicator defined in Chapter 4.

The pipe-bend finite-difference sensitivity verification in Chapter 6 is not assigned a separate configuration table. It uses the same geometry, physical parameters, wall-distance convention, penalty settings, objective, filter, and solver setup as the pipe-bend topology optimization run. The only additional settings are the finite-difference sampling parameters listed in Table B.3.

Table B.1: Shared wall-distance and SA topology-penalty conventions for the Alexandersen cases.

| **Parameter** | **Setting** |
|---|---|
| SA wall-distance mode | `reciprocal_penalized` |
| SA wall-density source | `design` |
| Solid threshold for wall source | $\bar{\gamma} = 0.50$ |
| SA working-variable penalty | Power-law solid indicator with amplitude $10^3$ and exponent $n = 4$ |
| Wall-distance penalty | Power-law solid indicator with amplitude $10^3$ and exponent $n = 4$ |
| Wall-distance recovery | Reciprocal recovery from the relaxed Eikonal variable $G$ |
| SA inlet turbulence estimate | $I = 7.5\%$, $\ell/H = 0.05$, eddy-viscosity ratio capped at 50 |
| Optimization objective | Average inlet pressure with fixed outlet pressure $p = 0$ |

## B.3 Alexandersen Pipe Bend

The pipe-bend case follows the setup of Bayat et al. [11]. The reference paper uses a standard $k$-$\varepsilon$ model with wall functions; the present thesis uses the wall-resolved SA implementation and evaluates the $V_f = 0.25$ design.

Table B.2: Geometric, physical, and boundary-condition parameters for the Alexandersen pipe-bend case.

| **Parameter** | **Setting** |
|---|---|
| Reference case | Pipe-bend benchmark from Bayat et al. [11] |
| Reference figure for setup | Bayat et al. Fig. 10 |
| Reference velocity figure | Bayat et al. Fig. 15 |
| Reference turbulence model | Standard $k$-$\varepsilon$ with wall functions |
| Turbulence model in this thesis | Spalart-Allmaras |
| Design-domain dimensions | Square design region $L \times L$ with $L = 1$ |
| Full computational domain | $x/L \in [-0.2, 1]$, $y/L \in [-0.2, 1]$ |
| Non-design inlet duct | Length $0.2L$, height $0.2L$, inlet over $y/L \in [0.7, 0.9]$ |
| Non-design outlet duct | Width $0.2L$, length $0.2L$, outlet over $x/L \in [0.7, 0.9]$ on the lower boundary |
| Volume fraction | $V_f = 0.25$ |
| Reynolds number | $Re = 5{,}000$, based on $U_{\text{in}} H_{\text{in}} \rho/\mu$ with $H_{\text{in}} = 0.2L$ |
| Density and viscosity | $\rho = 1.0$, $\mu = 4.0 \times 10^{-5}$ |
| Inlet condition | Uniform inlet velocity $\mathbf{u} = (1, 0)$; SA inlet estimated from the shared turbulence-intensity settings in Table B.1 |
| Outlet condition | Pressure outlet $p = 0$ on the lower outlet, with the normal/-tangential component constraint used by the FEniCS configuration |
| Wall conditions | No-slip velocity and $\tilde{\nu} = 0$ on external walls; topology-created solids enter through the design-density penalties |
| Validation metrics | Average inlet pressure, pressure drop, $\Phi_D$, and relative changes between FEniCS-SA, Ansys-SA, and Ansys-SST |

TABLE B.3: Numerical FEniCS solver, topology-optimization, and sensitivity-verification parameters for the Alexandersen pipe bend.

| Parameter | Symbol / option | Setting |
|---|---|---|
| Flow finite elements | $(\mathbf{u}, p)$ | Taylor–Hood $P_2/P_1$ continuous Galerkin elements |
| SA working-variable space | $\tilde{\nu}$ | $P_1$ continuous Galerkin |
| Design-density space | $\gamma$ | $DG_0$ cellwise constant density |
| hh | – | `mesh_yplus1.xdmf`; 39,128 vertices, 77,334 triangles, $h_{\max} = 0.007L$ |
| Objective | $J_p$ | Average inlet pressure |
| Initial design density | $\gamma_0$ | $V_f = 0.25$, matched to the filtered volume |
| Brinkman penalty | $\alpha(\bar{\gamma})$ | RAMP form with $\alpha_{\text{fluid}} = 0$, $\alpha_{\text{solid}} = 100$ |
| RAMP continuation | $q$ | Code schedule $q = 1/q_\alpha = [1/150,\ 1/75,\ 1/30,\ 1/15]$ |
| Projection continuation | $\beta$ | $[4,\ 6,\ 9,\ 13]$ with threshold $\eta = 0.50$ |
| MMA move limits | – | $[0.05,\ 0.04,\ 0.03,\ 0.02]$ |
| Maximum optimization iterations | – | Four continuation stages with 25 MMA iterations each; stop if objective change is below $10^{-5}$ for five iterations |
| Filter radius | $r_f$ | $4h_{\max} = 0.028L$ |
| Forward flow solver | – | IPCS flow solves for frozen-SA Picard updates, followed by a warm-started SNES Newton line-search final flow solve |
| Frozen NS-SA coupling | – | Three Picard updates per design iteration with SA relaxation factor 0.20 |
| Linear solver | – | MUMPS for SNES/state and adjoint solves; BiCGSTAB with ILU preconditioning for IPCS warm-start solves |
| Adjoint treatment | – | Frozen-turbulence adjoint in $(\mathbf{u}^*, p^*)$; $\tilde{\nu}$ and $G$ are frozen during gradient assembly |
| Sensitivity verification samples | – | Five active DG0 cells, seed 13 |
| Sensitivity verification step | $\Delta h$ | $10^{-6}$, central finite difference |
| Sensitivity verification fields held fixed | – | SA working variable $\tilde{\nu}^0$ and reciprocal wall-distance field $G^0$ |

Table B.4: Ansys Fluent mesh-independence table for the extracted turbulent Alexandersen pipe-bend geometry.

| **Parameter** | **Coarse** | **Medium** | **Fine** |
|---|---|---|---|
| Bulk mesh type | [TBD] | [TBD] | [TBD] |
| Near-wall treatment | [TBD] | [TBD] | [TBD] |
| First-layer height $\Delta y_1$ [m] | [TBD] | [TBD] | [TBD] |
| Inflation layers | [TBD] | [TBD] | [TBD] |
| Growth rate | [TBD] | [TBD] | [TBD] |
| Elements / cells | [TBD] | [TBD] | [TBD] |
| Maximum $y^+$, SA | [TBD] | [TBD] | [TBD] |
| Maximum $y^+$, SST $k$-$\omega$ | [TBD] | [TBD] | [TBD] |
| $\Phi_D^{\mathrm{SA}}$ [W/m] | [TBD] | [TBD] | [TBD] |
| $\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ [W/m] | [TBD] | [TBD] | [TBD] |
| $\Delta p^{\mathrm{SA}}$ [Pa] | [TBD] | [TBD] | [TBD] |
| $\Delta p^{\mathrm{SST}\ k\text{-}\omega}$ [Pa] | [TBD] | [TBD] | [TBD] |
| Relative change, Medium–Fine | | [TBD]% | |
| Selected validation mesh | | [TBD] | |

## B.4 Alexandersen U-Bend

The U-bend case follows Fig. 11 and Table 4 of Bayat et al. [11]. It uses the same turbulent topology optimization framework as the pipe bend, but forces a 180° return path around a fixed internal separator.

TABLE B.5: Geometric, physical, and boundary-condition parameters for the Alexandersen U-bend case.

| **Parameter** | **Setting** |
|---|---|
| Reference case | U-bend benchmark from Bayat et al. [11] |
| Reference figure/table for setup | Bayat et al. Fig. 11 and Table 4 |
| Reference turbulence model | Standard $k$-$\varepsilon$ with wall functions |
| Turbulence model in this thesis | Spalart-Allmaras |
| Design-domain dimensions | Square design region $L \times L$ with $L = 1$ |
| Full computational domain | $x/L \in [-0.2, 1]$, $y/L \in [0, 1]$ |
| Port geometry | Two left-side ports of height $0.2L$; inlet over $y/L \in [0.55, 0.75]$, outlet over $y/L \in [0.25, 0.45]$ |
| Fixed separator | Bar thickness $0.10L$, rounded-tip radius $0.05L$, extending from $x/L = -0.2$ to tip at $x/L = 0.5$ |
| Volume fraction | $V_f = 0.27$ |
| Reynolds number | $Re = 5{,}000$, based on $U_{\text{in}} H_{\text{port}} \rho/\mu$ with $H_{\text{port}} = 0.2L$ |
| Density and viscosity | $\rho = 1.0$, $\mu = 4.0 \times 10^{-5}$ |
| Inlet condition | Uniform inlet velocity $\mathbf{u} = (1, 0)$; SA inlet estimated from the shared turbulence-intensity settings in Table B.1 |
| Outlet condition | Pressure outlet $p = 0$ on the lower left port, with the component constraint used by the FEniCS configuration |
| Wall conditions | No-slip velocity and $\tilde{\nu} = 0$ on external walls and the fixed separator; topology-created solids enter through the design-density penalties |
| Validation metrics | Average inlet pressure, pressure drop, $\Phi_D$, and relative changes between FEniCS-SA, Ansys-SA, and Ansys-SST |

Table B.6: Numerical FEniCS solver and topology-optimization parameters for the Alexandersen U-bend.

| Parameter | Symbol / option | Setting |
|---|---|---|
| Flow finite elements | $(\mathbf{u}, p)$ | Taylor–Hood $P_2/P_1$ continuous Galerkin elements |
| SA working-variable space | $\tilde{\nu}$ | $P_1$ continuous Galerkin |
| Design-density space | $\gamma$ | $DG_0$ cellwise constant density |
| Reference FEniCS mesh | – | `mesh_yplus1.xdmf`; wall-resolved triangular Gmsh mesh with 45,280 vertices, 85,780 triangles, near-wall refinement based on $y^+ \approx 1$, and far-field $h_{\max} = 0.007L$ |
| Objective | $J_p$ | Average inlet pressure |
| Initial design density | $\gamma_0$ | $V_f = 0.27$, matched to the filtered volume |
| Brinkman penalty | $\alpha(\bar{\gamma})$ | RAMP form with $\alpha_{\text{fluid}} = 0$, $\alpha_{\text{solid}} = 100$ |
| RAMP continuation | $q$ | Code schedule $q = 1/q_\alpha = [1/150,\ 1/75,\ 1/35,\ 1/12]$ |
| Projection continuation | $\beta$ | $[8,\ 10,\ 18,\ 18]$ with threshold $\eta = 0.50$ |
| MMA move limits | – | $[0.05,\ 0.04,\ 0.03,\ 0.02]$ |
| Maximum optimization iterations | – | Four continuation stages with 25 MMA iterations each; stop if objective change is below $10^{-5}$ for five iterations |
| Filter radius | $r_f$ | $4h_{\max} = 0.028L$ |
| Forward flow solver | – | IPCS flow solves for frozen-SA Picard updates, followed by a warm-started SNES Newton line-search final flow solve |
| Frozen NS-SA coupling | – | Three Picard updates per design iteration with SA relaxation factor 0.20 |
| Linear solver | – | MUMPS for SNES/state and adjoint solves; BiCGSTAB with ILU preconditioning for IPCS warm-start solves |
| Adjoint treatment | – | Frozen-turbulence adjoint in $(\mathbf{u}^*, p^*)$; $\tilde{\nu}$ and $G$ are frozen during gradient assembly |

TABLE B.7: Ansys Fluent mesh-independence table for the extracted turbulent Alexandersen U-bend geometry.

| **Parameter** | **Coarse** | **Medium** | **Fine** |
|---|---|---|---|
| Bulk mesh type | [TBD] | [TBD] | [TBD] |
| Near-wall treatment | [TBD] | [TBD] | [TBD] |
| First-layer height $\Delta y_1$ [m] | [TBD] | [TBD] | [TBD] |
| Inflation layers | [TBD] | [TBD] | [TBD] |
| Growth rate | [TBD] | [TBD] | [TBD] |
| Elements / cells | [TBD] | [TBD] | [TBD] |
| Maximum $y^+$, SA | [TBD] | [TBD] | [TBD] |
| Maximum $y^+$, SST $k$-$\omega$ | [TBD] | [TBD] | [TBD] |
| $\Phi_D^{\mathrm{SA}}$ [W/m] | [TBD] | [TBD] | [TBD] |
| $\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ [W/m] | [TBD] | [TBD] | [TBD] |
| $\Delta p^{\mathrm{SA}}$ [Pa] | [TBD] | [TBD] | [TBD] |
| $\Delta p^{\mathrm{SST}\ k\text{-}\omega}$ [Pa] | [TBD] | [TBD] | [TBD] |
| Relative change, Medium–Fine | | [TBD]% | |
| Selected validation mesh | | [TBD] | |

# Appendix C

# Performance Comparison of Laminar and Turbulent Benchmark Topologies

This appendix supports the results in Chapter 6 by comparing topologies optimized with a laminar formulation against topologies optimized with the turbulent SA formulation. The comparison is performed for the pipe-bend and U-bend benchmarks from Bayat et al. [11].

## C.1 Methodology and Evaluation Pipeline

Each benchmark is solved twice. The laminar reference design is optimized at $Re = 1$, where the optimizer primarily responds to creeping-flow viscous losses. The turbulent design is optimized at the target high-Reynolds operating condition using the SA topology optimization framework described in Chapter 4. Apart from replacing the SA turbulent state equations by the laminar flow formulation and setting $Re = 1$, the laminar topology-optimization runs use the same benchmark-specific FEniCS setup as the turbulent Alexandersen runs: identical mesh, design-domain restrictions, non-design regions, volume constraint, uniform prescribed inlet velocity profile, pressure outlet, inlet-pressure objective, Brinkman penalty limits, filter radius, projection threshold, continuation schedules, MMA move limits, and maximum MMA iterations. This keeps the laminar-vs-turbulent comparison focused on the flow-physics model rather than on optimization or boundary-condition choices.

Because these FEniCS physical and numerical settings are inherited directly from the turbulent benchmark definitions in Appendix B, they are not repeated as separate configuration tables in this appendix. The additional tables reported here are therefore limited to the Ansys mesh-independence studies for the extracted laminar geometries and the final CFD metric comparisons against the turbulent-design results from Chapter 6.

After optimization, both projected density fields are thresholded and extracted as sharp geometries. The extracted laminar and turbulent designs are then evaluated in Ansys Fluent under the same turbulent operating condition. The key performance metrics are viscous dissipation and pressure drop, reported with both SA and SST $k$-$\omega$ turbulence models when available. The purpose of this appendix is not merely to show that the topologies look different, but to test whether the turbulent topology actually performs better when both designs are exposed to the same turbulent flow.

For fairness, the extracted laminar geometries receive their own Ansys mesh-independence checks. The turbulent-design grid studies are reported in Table B.4 of Section B.3 and Table B.7 of Section B.4 in Appendix B; this appendix adds the corresponding laminar-design mesh tables and final laminar-vs-turbulent performance comparisons.

## C.2 Pipe-Bend Benchmark

The pipe bend tests whether a topology optimized under turbulent conditions produces a more separation-aware 90° turning path than a laminar topology optimized at $Re = 1$. The turbulent case in Chapter 6 uses the $V_f = 0.25$ design.

### C.2.1 Topology Comparison

Figure C.1 compares the laminar and turbulent optimized topologies before sharp-geometry extraction. The expected distinction is that the laminar design can favor a shorter and more direct turning path, while the turbulent design should allocate fluid volume to reduce separation and momentum loss through the bend.

**Placeholder**

Projected density: laminar pipe-bend design ($Re = 1$)

(A) Laminar-optimized topology.

**Placeholder**

Projected density: turbulent pipe-bend design (SA)

(B) Turbulent-optimized topology.

FIGURE C.1: Laminar and turbulent topology comparison for the pipe-bend benchmark.

### C.2.2 Laminar-Design Mesh Independence

Table C.1 reports the Ansys mesh-independence study for the extracted laminar pipe-bend design. This table is separate from the turbulent-design mesh study in

**Placeholder**

Ansys turbulent velocity field through laminar pipe-bend design

(a) Laminar design evaluated turbulently.

**Placeholder**

Ansys turbulent velocity field through turbulent pipe-bend design

(b) Turbulent design evaluated turbulently.

Figure C.2: Velocity-magnitude comparison for the extracted pipe-bend designs under identical turbulent operating conditions.

Table B.4 of Section B.3 because the laminar and turbulent topologies generally produce different wall shapes and near-wall mesh requirements.

Table C.1: Ansys mesh-independence table for the extracted laminar pipe-bend design.

| **Parameter** | **Coarse** | **Medium** | **Fine** |
|---|---|---|---|
| Bulk mesh type | [TBD] | [TBD] | [TBD] |
| Near-wall treatment | [TBD] | [TBD] | [TBD] |
| First-layer height $\Delta y_1$ [m] | [TBD] | [TBD] | [TBD] |
| Inflation layers | [TBD] | [TBD] | [TBD] |
| Growth rate | [TBD] | [TBD] | [TBD] |
| Elements / cells | [TBD] | [TBD] | [TBD] |
| Maximum $y^+$, SA | [TBD] | [TBD] | [TBD] |
| Maximum $y^+$, SST $k$-$\omega$ | [TBD] | [TBD] | [TBD] |
| $\Phi_D^{\mathrm{SA}}$ [W/m] | [TBD] | [TBD] | [TBD] |
| $\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ [W/m] | [TBD] | [TBD] | [TBD] |
| $\Delta p^{\mathrm{SA}}$ [Pa] | [TBD] | [TBD] | [TBD] |
| $\Delta p^{\mathrm{SST}\ k\text{-}\omega}$ [Pa] | [TBD] | [TBD] | [TBD] |
| Relative change, Medium–Fine | | [TBD]% | |
| Selected comparison mesh | | [TBD] | |

### C.2.3 Performance Comparison

The final comparison in Table C.2 evaluates the extracted laminar and turbulent pipe-bend designs under identical turbulent operating conditions. A lower pressure drop and lower viscous dissipation for the turbulent design would support the need for a turbulent topology optimization framework rather than a laminar surrogate.

TABLE C.2: Performance comparison of the extracted pipe-bend topologies under identical turbulent operating conditions.

| **Design origin** | **Model** | $\Phi_D$ **[W/m]** | $\Delta p$ **[Pa]** | **Relative performance** |
|---|---|---|---|---|
| Laminar TO, $Re = 1$ | SA | [TBD] | [TBD] | Baseline |
| Turbulent TO, SA | SA | [TBD] | [TBD] | [TBD]% vs. laminar |
| Laminar TO, $Re = 1$ | SST $k$-$\omega$ | [TBD] | [TBD] | Baseline |
| Turbulent TO, SA | SST $k$-$\omega$ | [TBD] | [TBD] | [TBD]% vs. laminar |

## C.3 U-Bend Benchmark

The U-bend tests whether the turbulent optimizer remains advantageous when the flow must complete a 180° return path around a fixed separator. The turbulent case in Chapter 6 uses the $V_f = 0.27$ design.

### C.3.1 Topology Comparison

Figure C.3 compares the laminar and turbulent optimized U-bend topologies. The expected distinction is that the laminar design may create an overly tight return path around the separator tip, while the turbulent design should form a smoother passage that reduces separation and turbulent mixing losses.

**Placeholder**

Projected density: laminar U-bend design ($Re = 1$)

(A) Laminar-optimized topology.

**Placeholder**

Projected density: turbulent U-bend design (SA)

(B) Turbulent-optimized topology.

FIGURE C.3: Laminar and turbulent topology comparison for the U-bend benchmark.

### C.3.2 Laminar-Design Mesh Independence

Table C.3 reports the Ansys mesh-independence study for the extracted laminar U-bend design. The turbulent U-bend mesh study is reported separately in Table B.7 of Section B.4.

**Placeholder**

Ansys turbulent velocity field through laminar U-bend design

(A) Laminar design evaluated turbulently.

**Placeholder**

Ansys turbulent velocity field through turbulent U-bend design

(B) Turbulent design evaluated turbulently.

Figure C.4: Velocity-magnitude comparison for the extracted U-bend designs under identical turbulent operating conditions.

Table C.3: Ansys mesh-independence table for the extracted laminar U-bend design.

| **Parameter** | **Coarse** | **Medium** | **Fine** |
|---|---|---|---|
| Bulk mesh type | [TBD] | [TBD] | [TBD] |
| Near-wall treatment | [TBD] | [TBD] | [TBD] |
| First-layer height $\Delta y_1$ [m] | [TBD] | [TBD] | [TBD] |
| Inflation layers | [TBD] | [TBD] | [TBD] |
| Growth rate | [TBD] | [TBD] | [TBD] |
| Elements / cells | [TBD] | [TBD] | [TBD] |
| Maximum $y^+$, SA | [TBD] | [TBD] | [TBD] |
| Maximum $y^+$, SST $k$-$\omega$ | [TBD] | [TBD] | [TBD] |
| $\Phi_D^{\mathrm{SA}}$ [W/m] | [TBD] | [TBD] | [TBD] |
| $\Phi_D^{\mathrm{SST}\ k\text{-}\omega}$ [W/m] | [TBD] | [TBD] | [TBD] |
| $\Delta p^{\mathrm{SA}}$ [Pa] | [TBD] | [TBD] | [TBD] |
| $\Delta p^{\mathrm{SST}\ k\text{-}\omega}$ [Pa] | [TBD] | [TBD] | [TBD] |
| Relative change, Medium–Fine | | [TBD]% | |
| Selected comparison mesh | | [TBD] | |

### C.3.3 Performance Comparison

Table C.4 compares the extracted laminar and turbulent U-bend topologies under identical turbulent operating conditions. If the turbulent design lowers $\Delta p$ and $\Phi_D$ across both SA and SST $k$-$\omega$ evaluations, the result supports the thesis claim that high-Reynolds topology optimization should include turbulence directly in the design loop.

Table C.4: Performance comparison of the extracted U-bend topologies under identical turbulent operating conditions.

| **Design origin** | **Model** | $\Phi_D$ **[W/m]** | $\Delta p$ **[Pa]** | **Relative performance** |
|---|---|---|---|---|
| Laminar TO, $Re = 1$ | SA | [TBD] | [TBD] | Baseline |
| Turbulent TO, SA | SA | [TBD] | [TBD] | [TBD]% vs. laminar |
| Laminar TO, $Re = 1$ | SST $k$-$\omega$ | [TBD] | [TBD] | Baseline |
| Turbulent TO, SA | SST $k$-$\omega$ | [TBD] | [TBD] | [TBD]% vs. laminar |

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|---|---|---|
| **For generating new (research) ideas** | *add text here* | This use of GenAI is allowed except when it is prohibited by the teacher or the program of study. Correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI.<br><br>**Be careful:**<br><br>• Further verify in this case whether the idea is novel or not. It is likely that it is related to existing work. **If so, that existing work should be correctly referenced in the text** (without such reference you would be committing plagiarism and thus committing an irregularity). |
| **For generating synthetic data** | *add text here* | Use of GenAI for generating synthetic data is allowed, **provided that it is methodologically and ethically justifiable**, except when it is prohibited by the teacher or the program of study. Always correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI.<br><br>**Be careful:**<br><br>• Always **carefully evaluate** the generated synthetic data for quality and possible bias since the output is highly dependent on the quality of the data on which the models are trained. |

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| **For generating blocks of text (other than the allowed use without referencing mentioned above)** | | According to Article 84 of the Regulations on Education and Examinations your text should allow to correctly and properly assess your own knowledge, understanding and skills. **Therefore, inserting blocks of text without quotes and a reference to GenAI assistance in your work is not allowed.**<br>**Be careful:**<br>• If it is really needed to insert a block of text from a GenAI tool, for instance because of the nature of your assignment, mention it as a citation by using quotes and correctly mention the use of GenAI assistance in accordance with the instructions page on being transparent about the use of GenAI.<br>• However, in general, such GenAI use should be kept to an **absolute minimum**; you should always check the original sources. |
| **For generating visuals, video or audio** | *add text here* | Use of GenAI for generating visuals, audio or video is allowed except when it is prohibited by the teacher or the program of study.<br>**Be careful:**<br>• If used, refer to GenAI for visuals according to the usual referencing style, following the instructions on the page on referencing GenAI.<br>• If you work with existing visuals, audio or video, **that existing work should be correctly referenced in the text** (without such reference you would be committing plagiarism and thus an irregularity).<br>• Explain the usage in the methods section (if there is one) and optionally attach (or link to) the prompts with the full output (or history). |

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| **Other use (specify here; this may also include a combination of types of use mentioned above):** | *add text here* | To make sure other use of GenAI is allowed within the course or thesis, and if so, the conditions that may apply, contact the teaching staff of the course or the supervisor of the thesis **beforehand** and explain the intended GenAI use. Also inform the programme director.<br><br>Motivate how you would comply with Article 84 of the exam regulations. Explain the use and the added value of the AI tool you consider to use and how it is in accordance with the assignment or thesis and the KU Leuven guidelines on responsible use of Generative AI tools.<br><br>Depending on the kind of GenAI use, it may be needed to properly reference it in the text, in accordance with the instructions page on being transparent about the use of GenAI. |